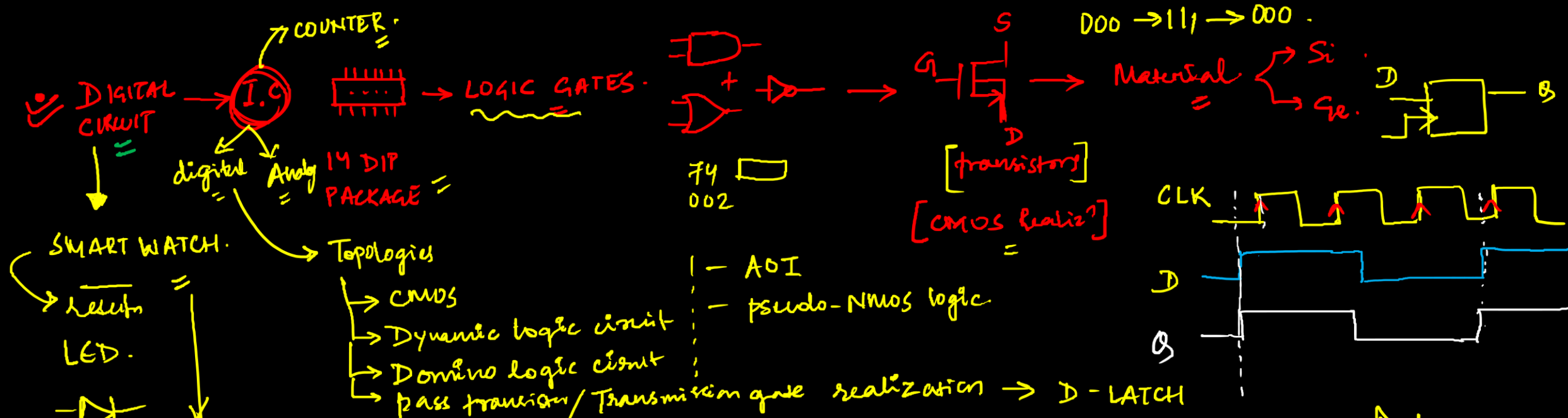




SEMICONDUCTOR PHYSICS

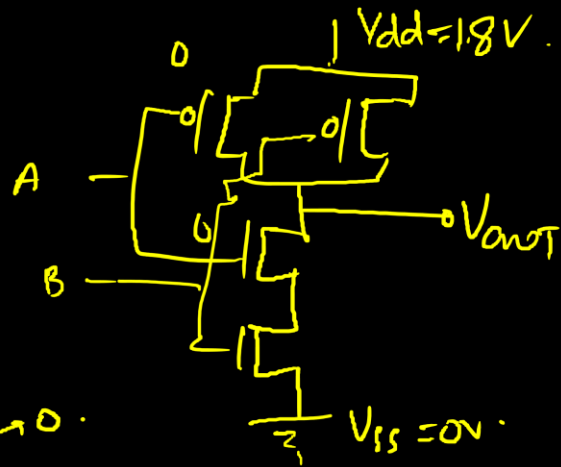
[02-02-2025]

Sailesh Mishra

RTL → synthesis (verification)
register transfer level

$$S = A \oplus B, C = A \cdot B$$

00 → 0
01 → 1 10 → 1 11 → 0

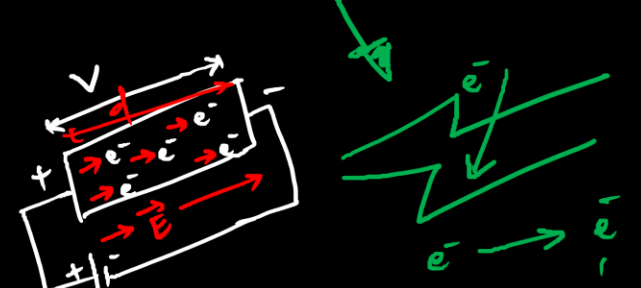
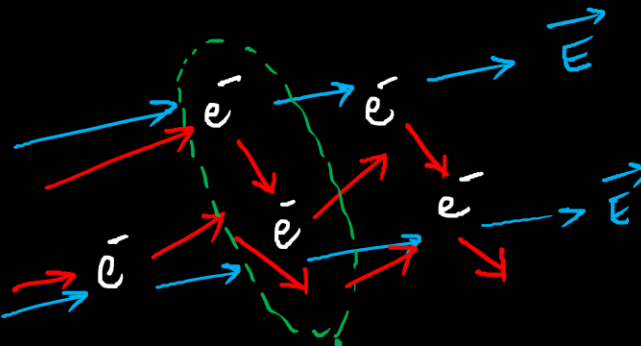


$$V_{supply} > V_T \neq 0mA$$

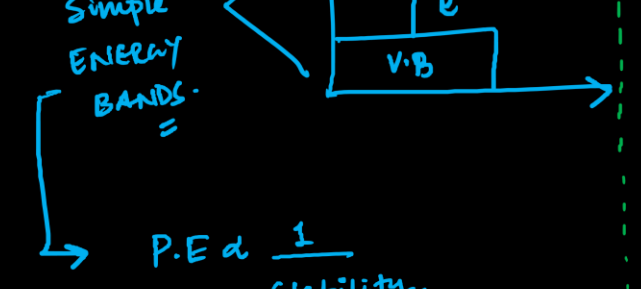
$0.2V < 0.6$

VLSI chts (logic chts)

- STATIC**: No CLK
- DYNAMIC**: has a clk signal.



$V_{DC} = 9V$
 $E = \frac{V}{d}$
 $F = qE$
 ELECTROSTATIC FORCE =
 Simple ENERGY BANDS =

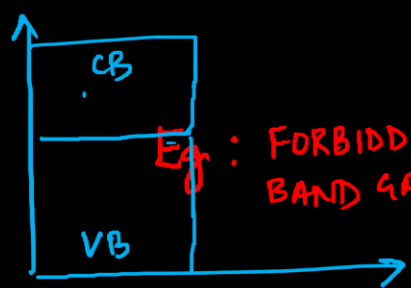


P.E $\propto \frac{1}{stability}$
 $TE = PE + KE$
 $= mgh + \frac{1}{2}mv^2$

MATERIALS

CONDUCTOR

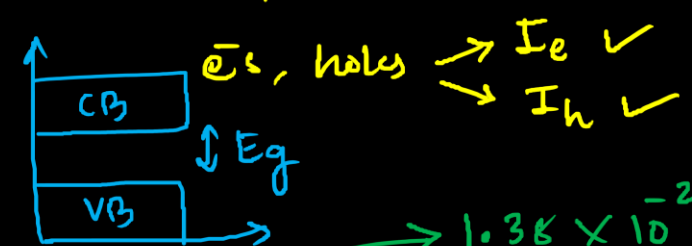
- uni polar Conduction
 $(e^-) \rightarrow I \checkmark$



REMARK:-
 The E_g in a Conductor is v. less !! So
 Easy Conduction.

SEMI-CONDUCTOR

- Bi-polar Conduction



E_g : FORBIDDEN BAND GAP ENERGY $\propto \frac{K_b \cdot T}{2e}$

$1.38 \times 10^{-23} JK^{-1}$
 Absolute Temp (K) $\rightarrow$ THERMAL ENERGY.
 [room temp]

$27^\circ C \rightarrow 273K + 27^\circ C = 300K$

Thermally Generated Current $\rightarrow I_e$
 $\rightarrow I_h$

$E_g = \frac{1.38 \times 10^{-23}}{1.6 \times 10^{-19}} \times \frac{1}{K} \times 300K$

$E_g = 25.87 \text{ eV} \approx 26 \text{ mV}$

$26 \times 10^{-3} = 0.026$

$0.026V$

Hence; It's NOT EASILY CONTROLLABLE or CALIBRABLE !!

INSULATOR

- NO CONDUCTION X

SEMI-CONDUCTOR

INSULATORS < Conductivity < CONDUCTORS

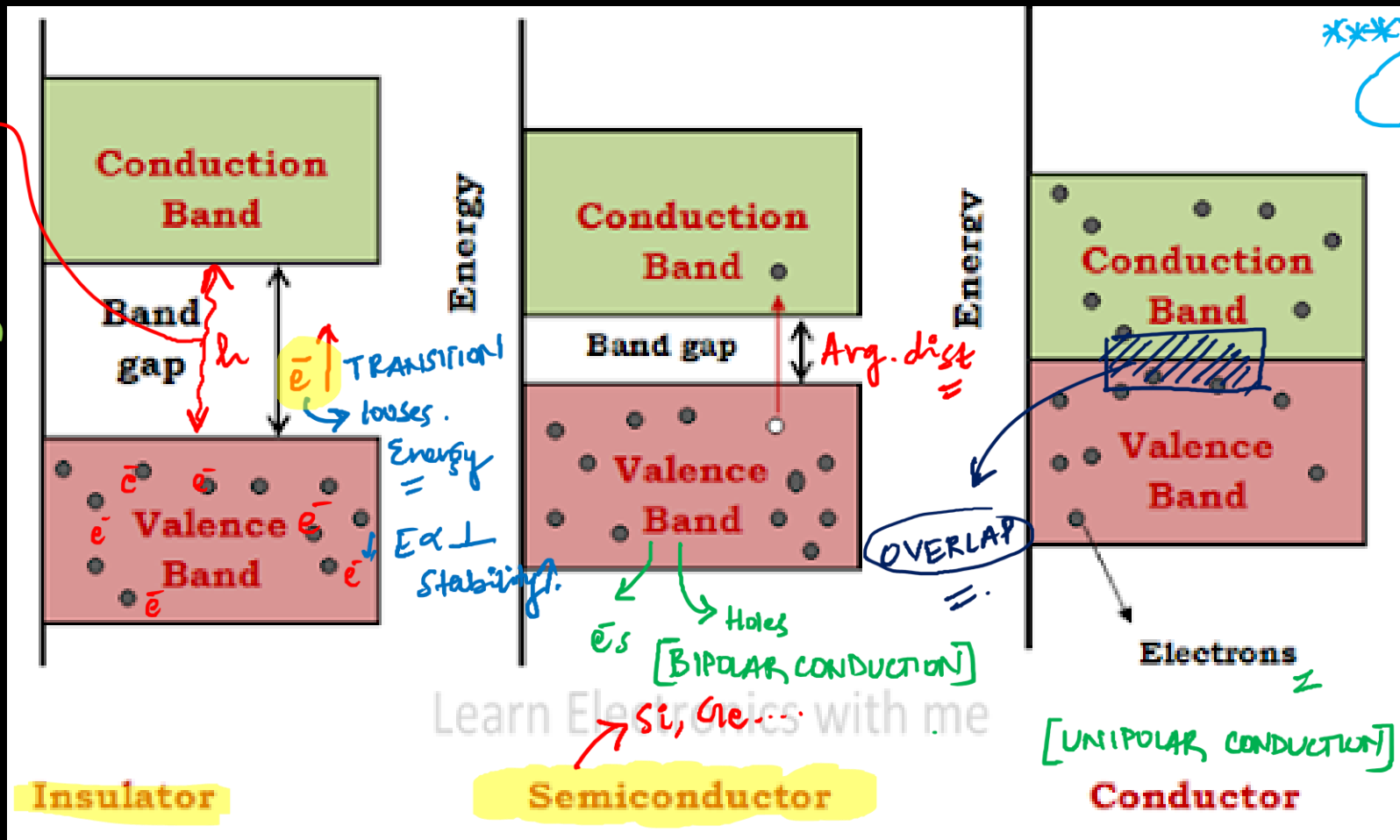
∴ This h is a distance, is a Material dependent property !!

of $h \uparrow$ PE $\uparrow$ TE $\uparrow$ (E $\uparrow$, stability $\downarrow$)

dist $\uparrow$ E $\uparrow$ Cond. $\downarrow$

$$E_g = \frac{1.38 \times 10^{-24} \text{ J/K} \times 300 \text{ K}}{1.6 \times 10^{-19}}$$

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DRIFT VELOCITY = $\frac{1}{2} m v_d^2$

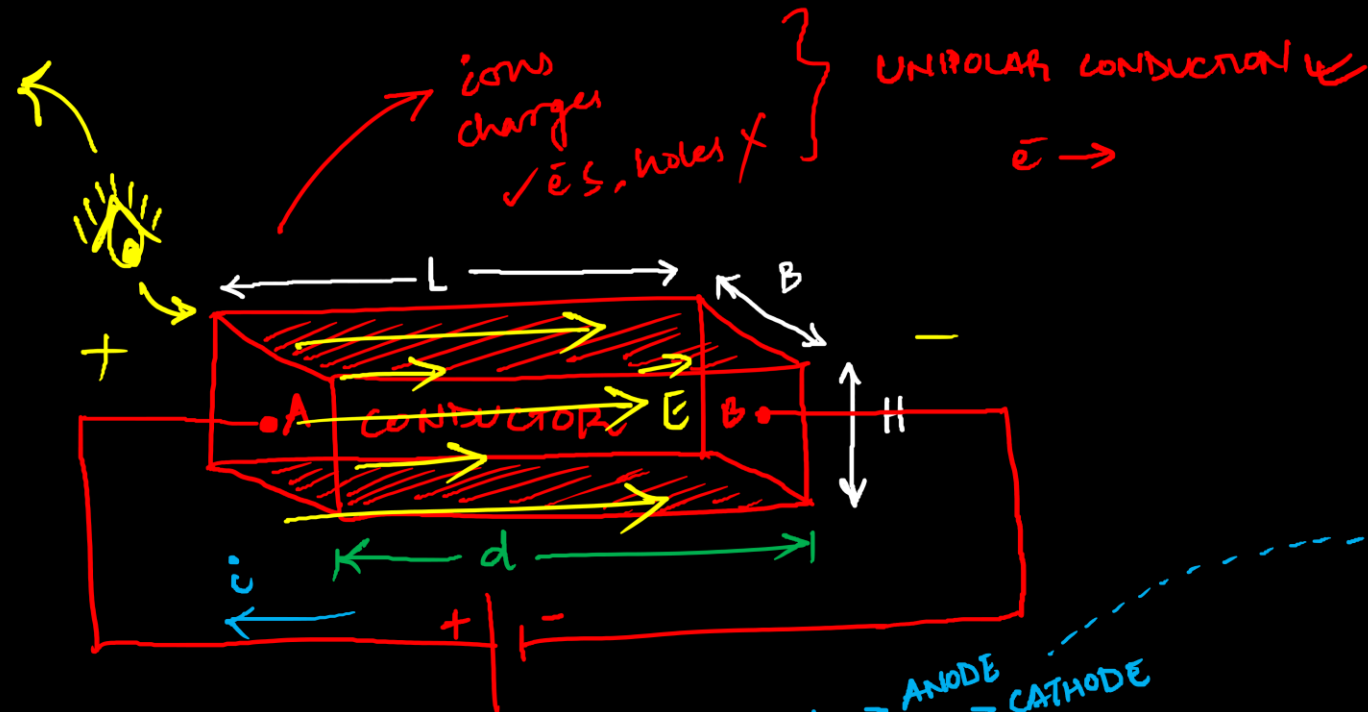
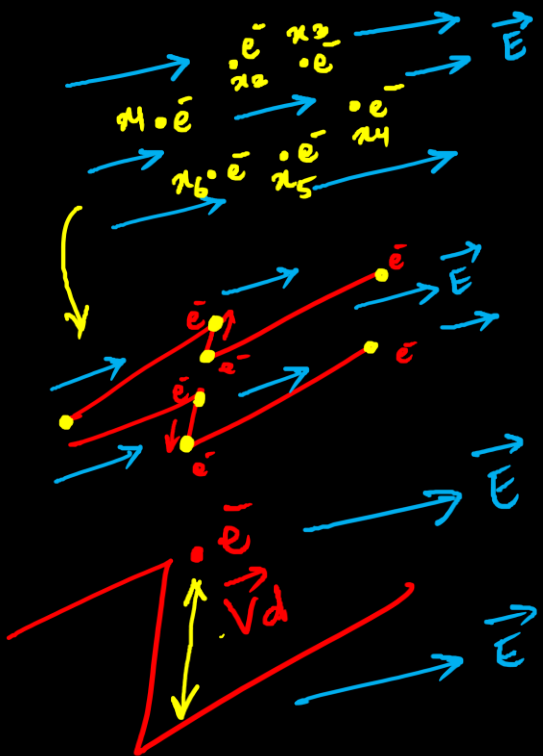
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[Band Gap Energy level]

It is directly proportional to $\frac{k_B T}{2}$

where $K_b \rightarrow$ Boltzman Constant, $T =$ Absolute Temp. = $27^\circ\text{C} = 273 + 27^\circ\text{C} = 300\text{K}$.
 $= 1.38 \times 10^{-24} \text{ J/K}$ $2 =$ charge of $e^- = 1.6 \times 10^{-19} \text{ C}$

Learn Electronics with me

Si, Ge...



"9V Battery"



$$\vec{F} = q\vec{E} = 1.6 \times 10^{-19} \text{ C} \times 31.4 \text{ N/Cm}^2$$

$$V_{DC} = V_{AB} = V_A - V_B : \text{Relative Potential difference}$$

$$= 9 - 0 \text{ V}$$

$$= 9 \text{ V}$$

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 $\vec{E} = \frac{V}{d}$
 emf (potential): ENERGY $\uparrow$
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$\rightarrow$ V-I Characteristics.
 $\rightarrow$ due to emf; We're getting a $\neq \vec{E}$.

$$18 - 9 = 9 \text{ V} !!$$

ANODE $\rightarrow$ CATHODE

Electrolytes
 [ions, charges].
 $=$
 emf (V) $\rightarrow \vec{E}$

$\odot \vec{E} = \text{ENERGY} = \frac{1}{2} \epsilon_0 \cdot E^2 = \text{Sum of } \underline{PE} + \underline{KE} = \underline{Mgh} + \frac{1}{2} m v_d^2 = m_e [gh + \frac{1}{2} \frac{V_d^2}{f}]$

$(9.1 \times 10^{-31} \text{ Kg})$

Environment dependent

permittivity of FREE SPACE

ϵ_0 : free space
or
Vacuum

$= 8.85 \times 10^{-12} \frac{C^2}{Nm^2}$

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Relative permittivity

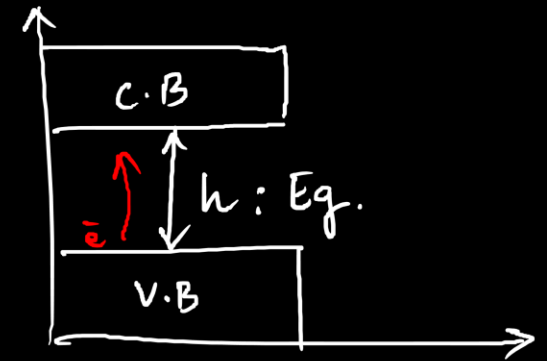
- Air = 1 ✓
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- G.C = 3-5 ✓
- H₂O = 79.6 ≈ 80 ✓

$\rightarrow \tau$ (Relaxation Time) $\frac{0}{0}$ It's the time B/W any 2 successive collisions of e^- s inside the conductor...

$\vec{E} \uparrow \quad V_d \uparrow$ (speed = $\frac{\text{dist}}{\text{time}}$) $\quad \tau \downarrow \rightarrow \tau$ (Relaxation Time).

$I = n A e V_d$

CONDUCTIVITY :- $[distance] \rightarrow Insulators > Semi-conductor > Conductors$
 (h)
Conductivity $\rightarrow Conductors > Semi-conductors > Insulators$
 (σ)



Why Semi-Conductors?

→ 0 EASIER CONTROLLABILITY. (Readily Available).

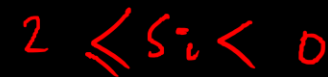
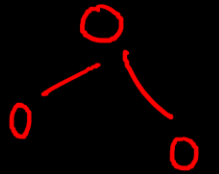
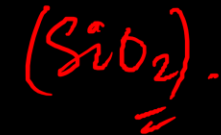
→ 0 LATTICE Structure.

→ 0 Thermally generated current.

→ 0 Cost Effective.

→ 0 Abundance In Availability.

$\rightarrow \hat{Si}$ (Sand): 2nd most Abundant Element/mineral found in Earth's crust.



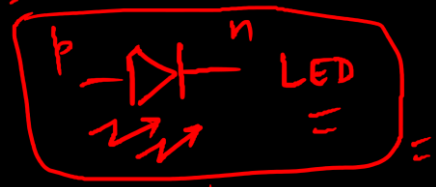
$\rightarrow$ Tetrahedral.

" $q = ne$ "

$i = \frac{dq}{dt} = \frac{d(ne)}{dt} = n \frac{dq}{dt} = \text{net}$

$n \uparrow \rightarrow i \uparrow$

TYPES OF SEMI-CONDUCTORS



INTRINSIC

→ pure form / Natural
 → These are NOT USED in VLSI Industry Mostly

Because they don't have good conductivity.

ELEMENTAL

Si, Ge

Group 14 elements

COMPOUND

GaAs, InP

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→ Intrinsic + DOPING !!

carriers / e^- ↑↑↑
 I_d ↑↑↑

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- Pentavalent impurities/dopants

- eg:- Sb, As, Bi, P
 Group 15 elements

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Trivalent impurities/dopants.

eg- Al, B.
 Group 13 elements

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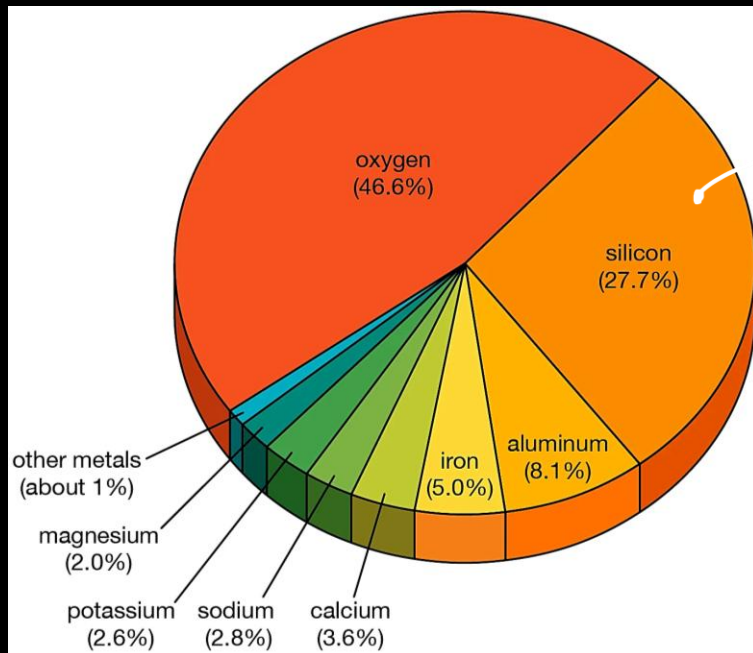
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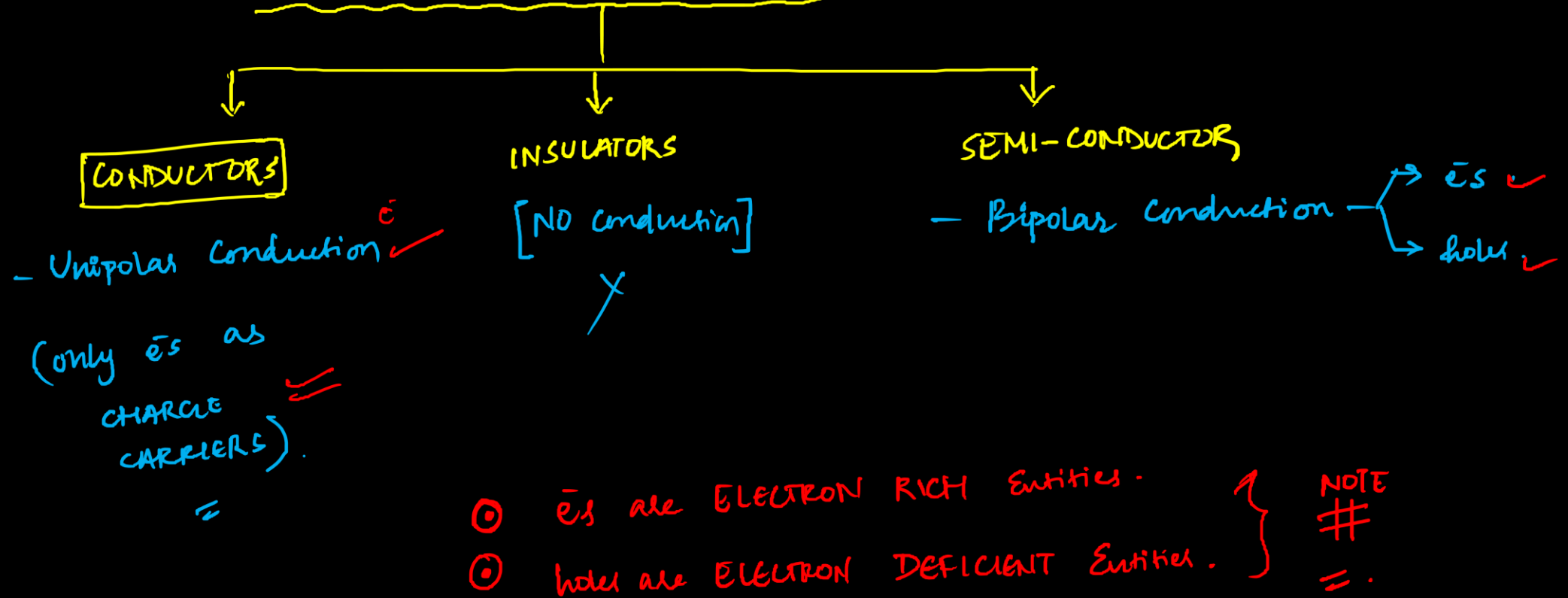


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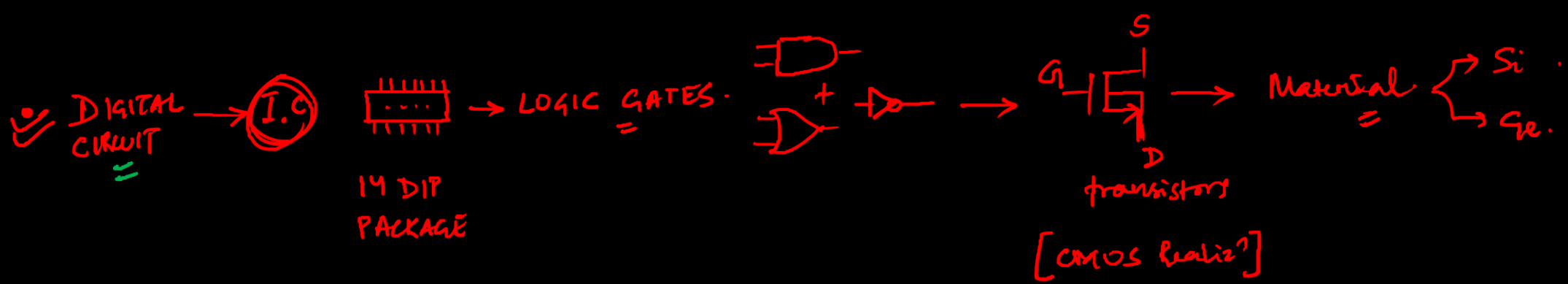


SEMICONDUCTOR PHYSICS

[RECAP]

[04-02-2025]

Sailesh Mishra



SEMICONDUCTOR PHYSICS

[02-02-2025]

Sailesh Mishra

SEMI-CONDUCTOR

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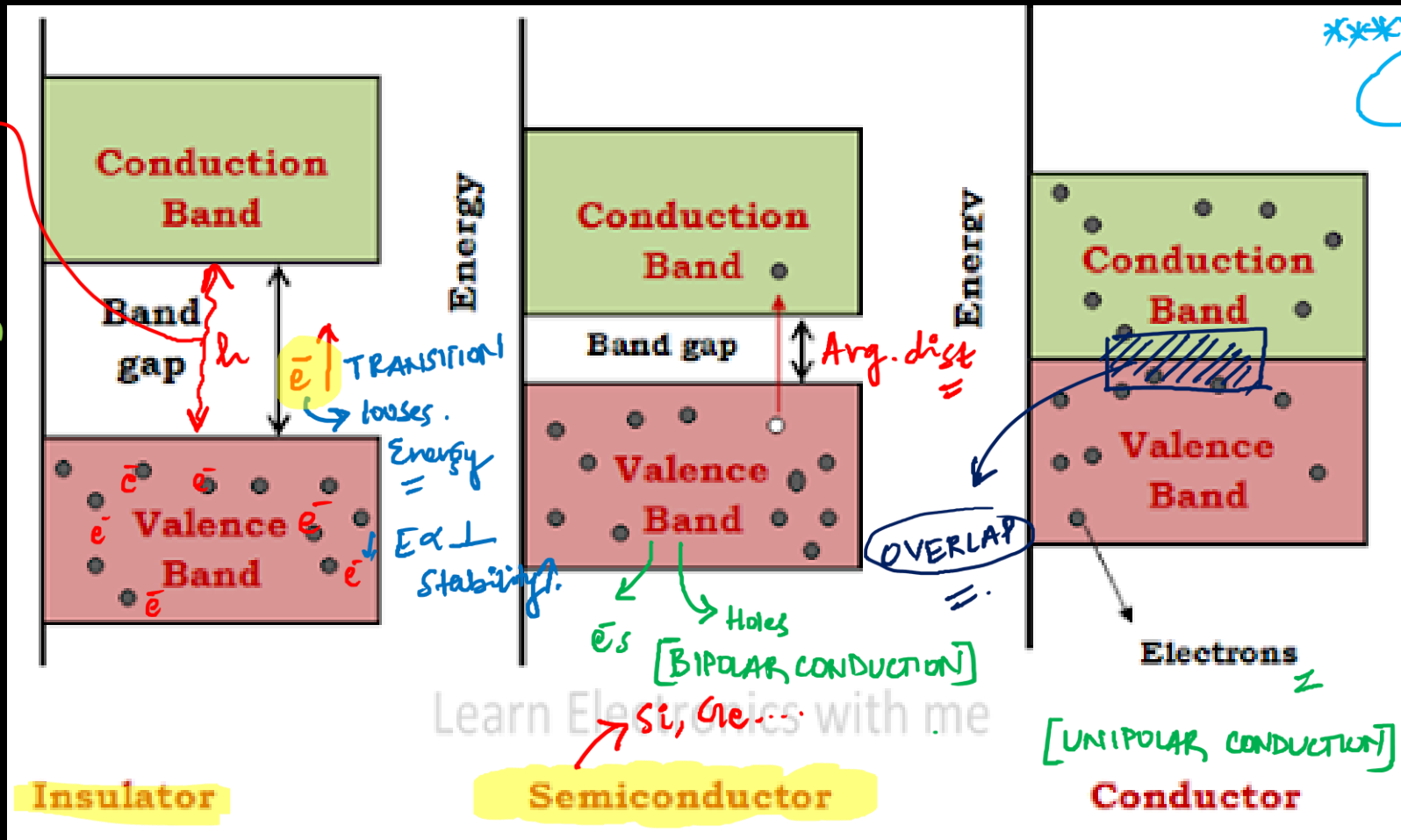
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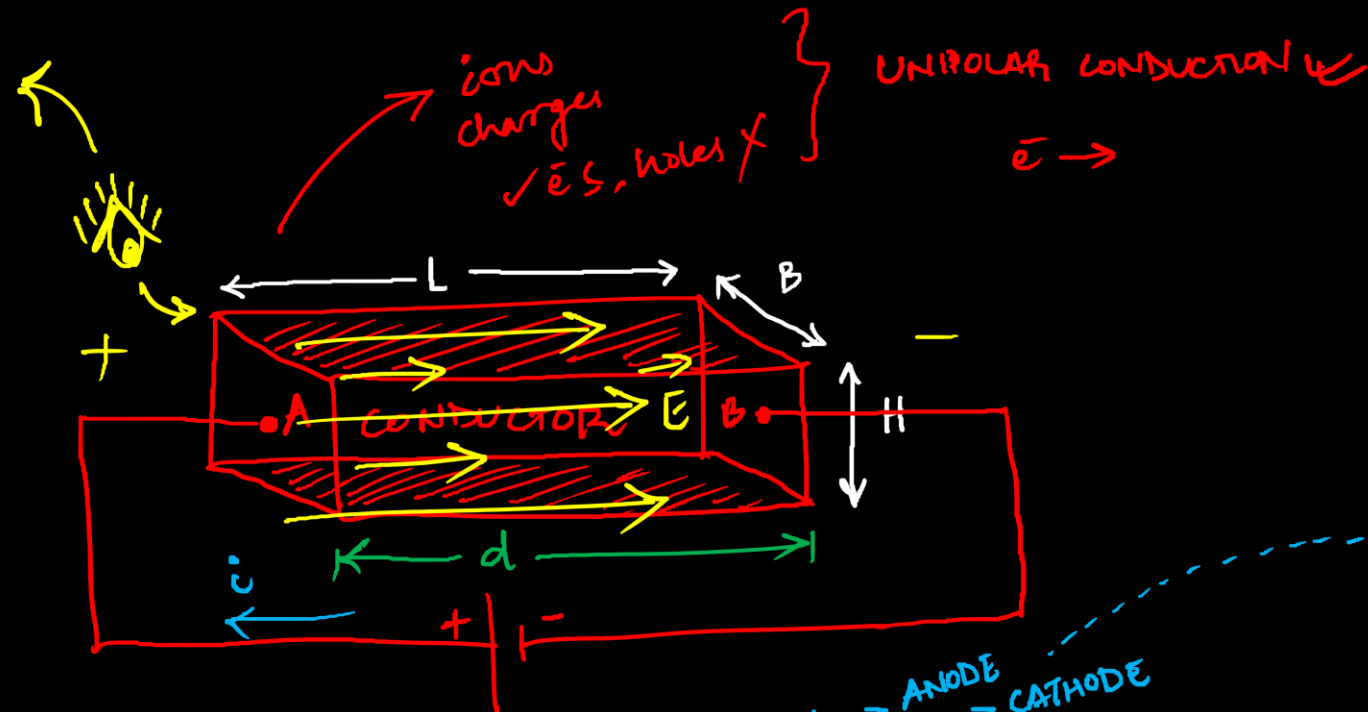
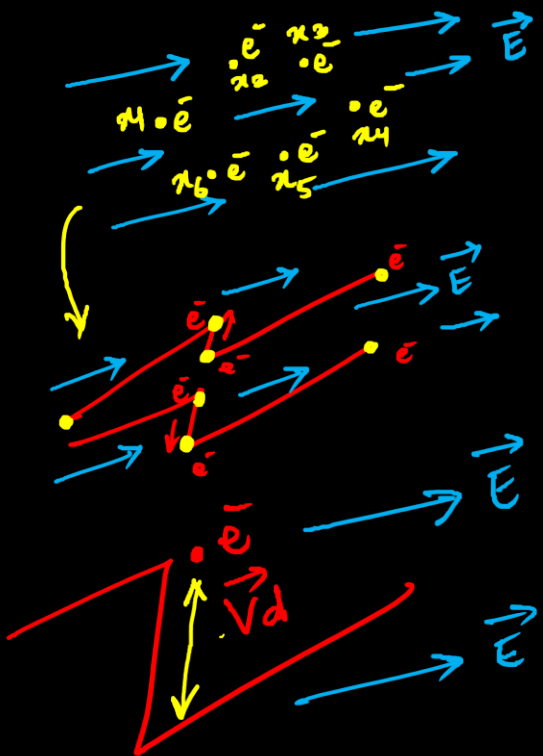
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$n \uparrow \rightarrow i \uparrow$

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→ pure form / Natural
 → These are NOT USED in VLSI Industry Mostly

Because they don't have good conductivity.

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Si, Ge

Group 14 elements

COMPOUND

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carriers / $e^- \uparrow \uparrow \uparrow$
 $I_d \uparrow \uparrow \uparrow$

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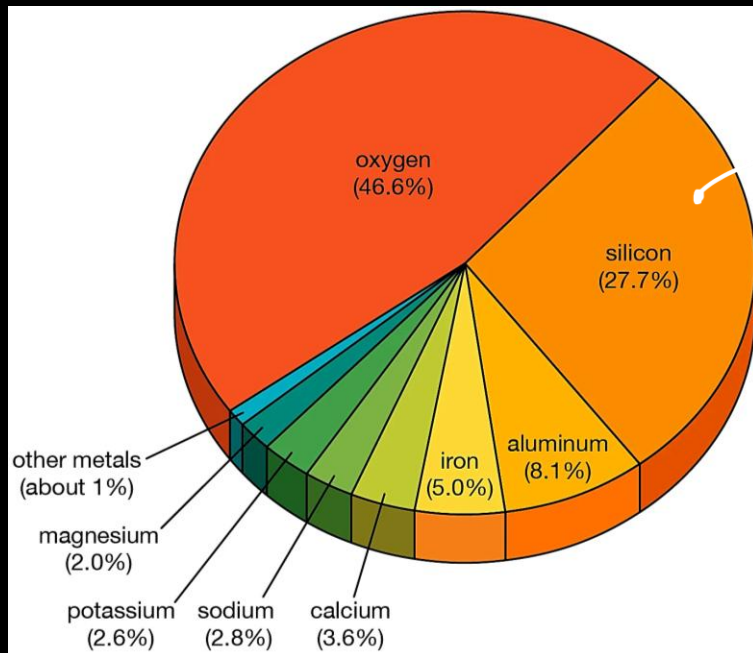
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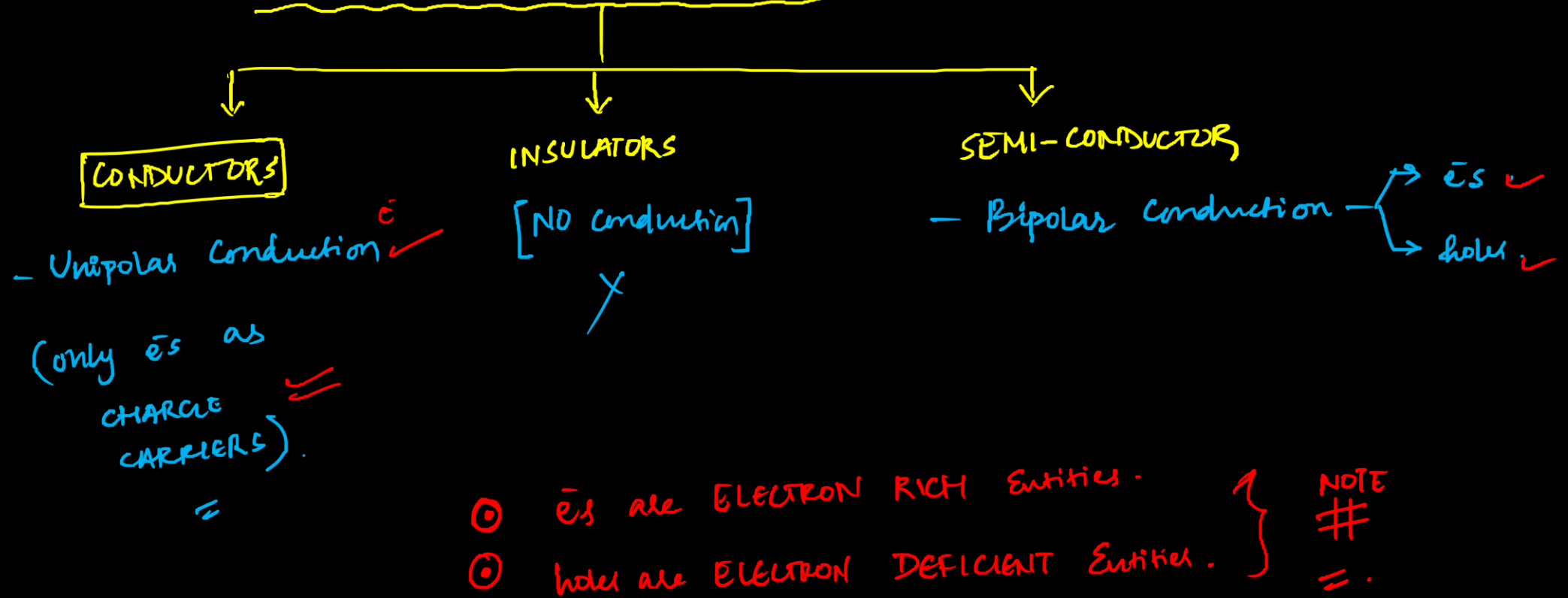


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SEMICONDUCTOR PHYSICS

[05-02-2025]

Sailesh Mishra

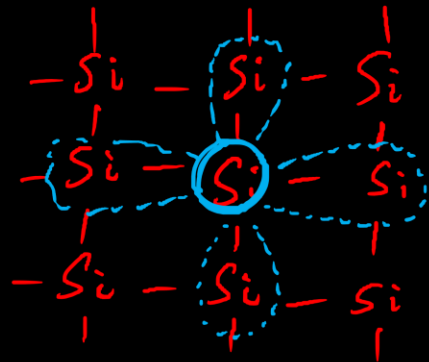
SEMI-CONDUCTORS

→ Si, Ge, GaAs, InP, N-TYPE and P-TYPE.

→ Group 14 Element-

→ Valency: 4 (Outermost shell 4 e⁻s).

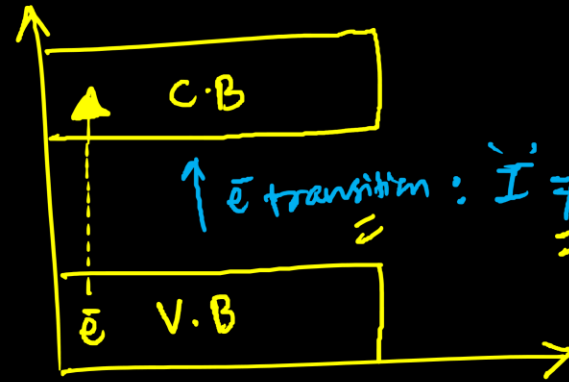
→ Tetravalent Element-



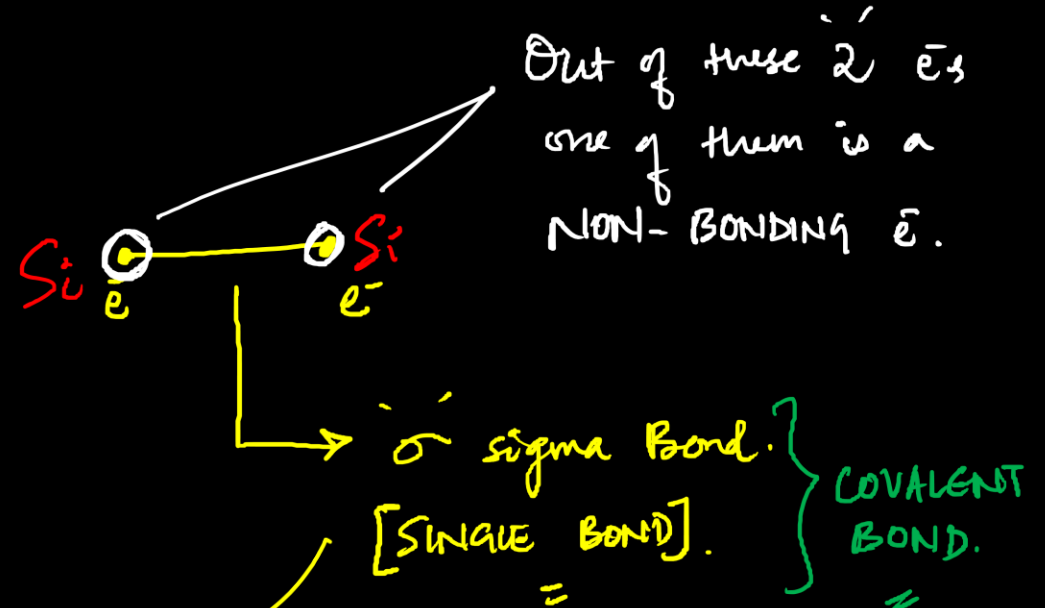
→ TETRAVALENT
ATOM

$$Si = 1.12 \text{ eV}$$

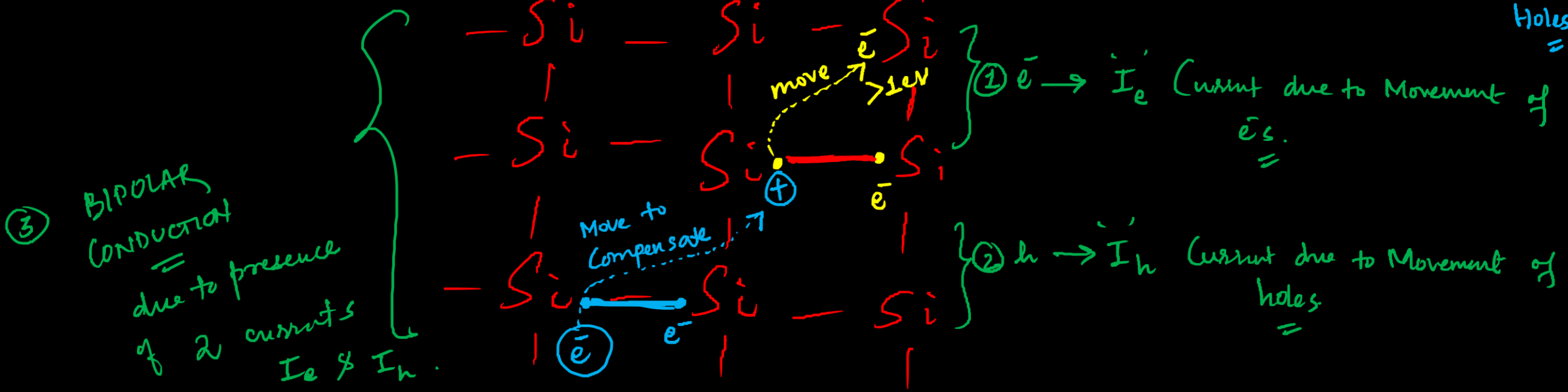
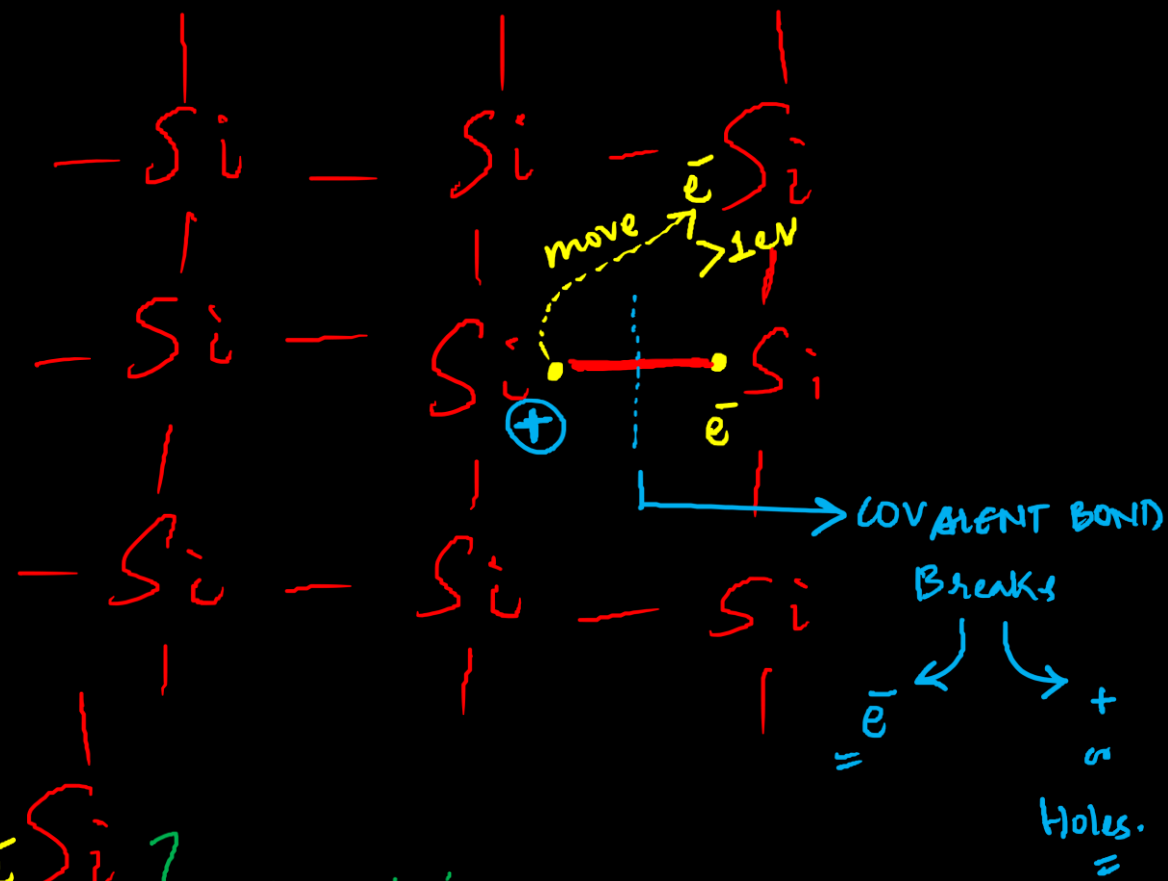
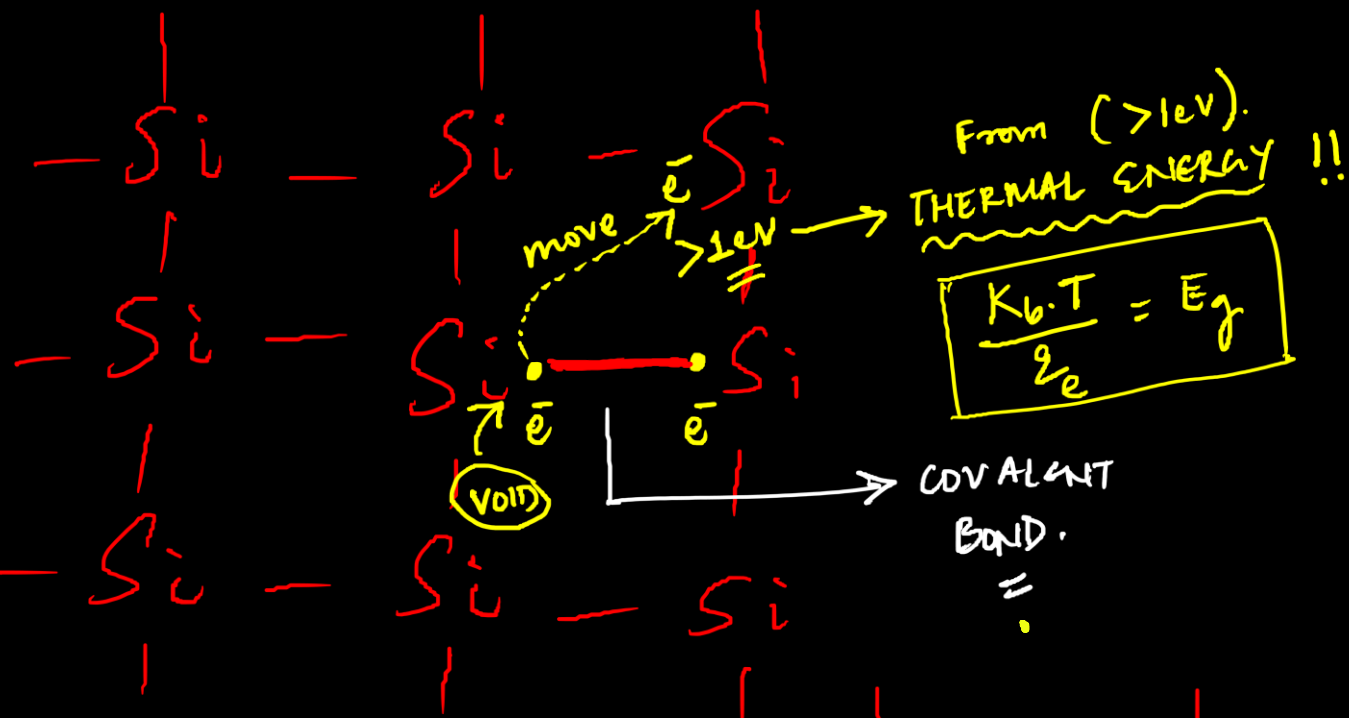
$$> 1 \text{ eV}$$



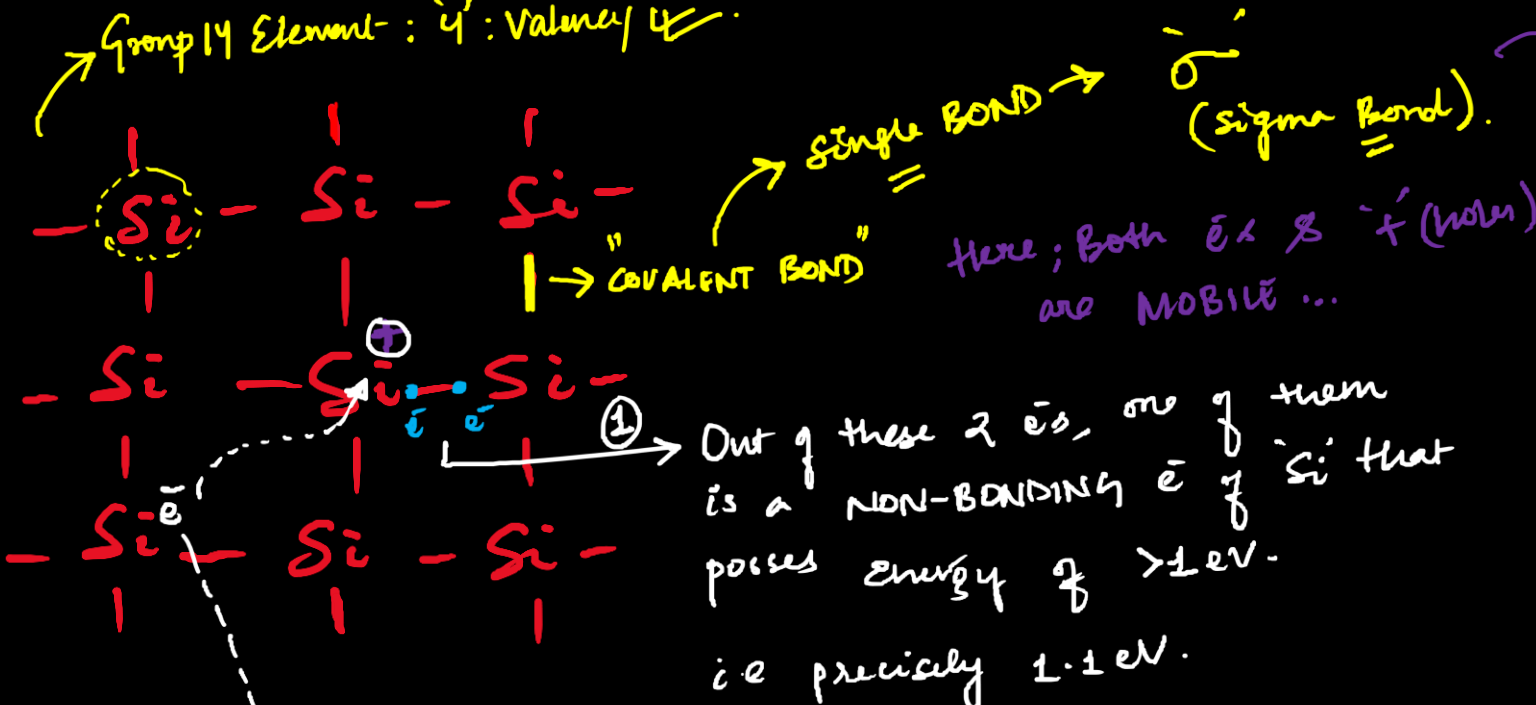
↑ e⁻ transition: $I \neq 0 \checkmark \rightarrow E_g \approx 25.87 \text{ mV}$
 $\approx \boxed{26 \text{ mV.}}$



• Mechanically Strong
 • Electrically Weak



Group 14 Element: 4 Valence/4



NOTE #

The Movement of e^- & h^+ charge direction is opposite to Each other

Hence; it's [BIPOLAR CONDUCTION].

* In CONDUCTORS; there is no h^+ @ holes, etc.

ONLY e^- ... hence [UNIPOLAR CONDUCTION]

Hence; It Moves from the lattice structure of Si $\rightarrow$ CB.

④ Now; e^- will move to fill that VACANCY...

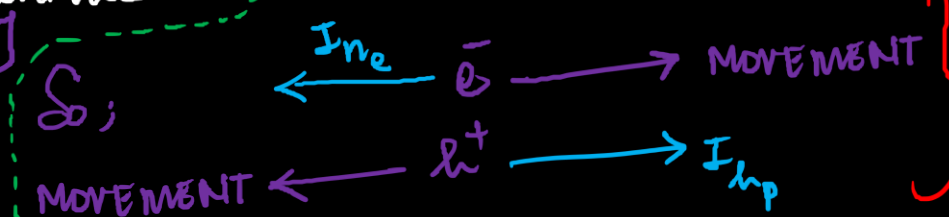
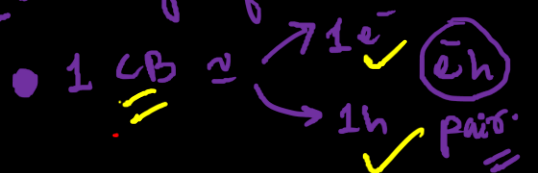
② Now; due to this; e^- DEFICIENT; Silicon Attains a $+ve$ charge on its head.

③ Now; to compensate this $[+ve \text{ charge}]$, Adjacent @ Neighbouring e^- come Nearby.

• Energy is nothing But Thermal Energy OR Vibrational Energy.

and; then this process will continue ...

[Breakage of COVALENT BONDS]

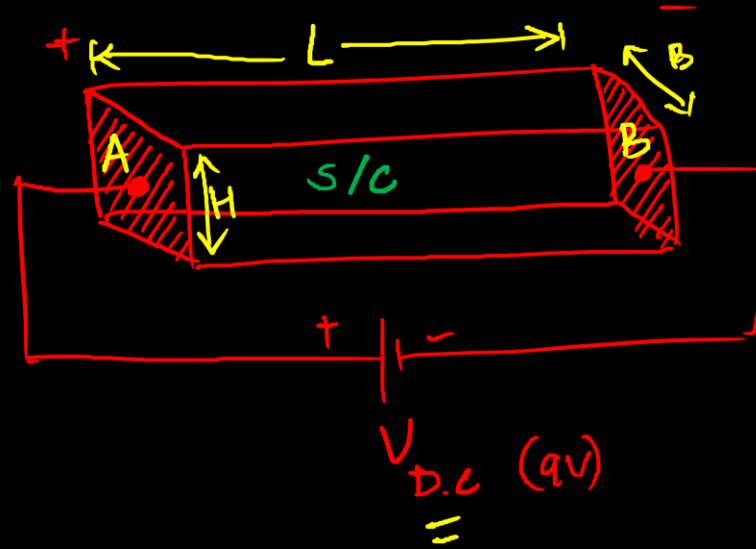


Net Current in Sustaining s/c. = $I = I_{n_e} + I_{h_p}$ ✓

Thermally Generated Current =

$$\sigma = n \cdot e \cdot \mu_e; \text{Conductors}$$

$$\sigma = n e \mu_h + n e \mu_e; \text{S/C}$$



$$R = \rho \cdot \frac{L}{A} = \frac{1}{\sigma} \cdot \frac{L}{A}$$

Conductors

$$R = \frac{1}{n e \mu_e} \cdot \frac{L}{A}$$

cond
e

S/C

$$R = \frac{1}{n e \mu_h + n e \mu_e} \cdot \frac{L}{A}$$

① μ = Mobility of $\bar{e}$ s or holes

$$= \frac{\vec{V}_d}{\vec{E}} = \frac{\text{drift velocity}}{\text{Electric field}}$$

$$\rightarrow I_{\text{total}} = I_e + I_h \quad **$$

So; $[CB \text{ breaking}] \rightleftharpoons e^- + e^- \text{ deficient site (HOLE)}$

Simultaneous process.

$e^- + h^+ \rightarrow \text{recombine} \rightarrow \text{BOND}$

This phenomenon is REVERSIBLE ...

when $T \uparrow$; Carrier conc. $\uparrow$

$[n = n_i = n_p]$

A material-dependent constant that incorporates factors like effective masses of electrons and holes, and the density of states in the conduction and valence bands. The value of A varies for different semiconductor materials.

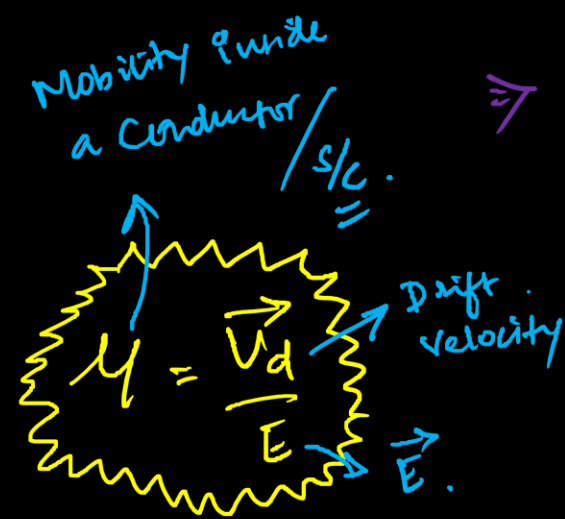
Intrinsic Carrier conc. $n_i = A \cdot T^{3/2} \cdot e^{-\frac{E_g}{2KT}}$

$\rightarrow$ Forbidden Band gap energy E_g

$\rightarrow$ Absolute Temp T

$\rightarrow$ [In KELVIN Scale]

$n_i^2 = A^2 \cdot T^3$



At a Given const. Temperature ...

$n_p = n_i^2 = \text{const} \dots [LAW \text{ OF MASS ACTION}]$

$\rightarrow 1.45 \times 10^{10} \text{ cm}^{-3}$

(2) For Semi-conductors

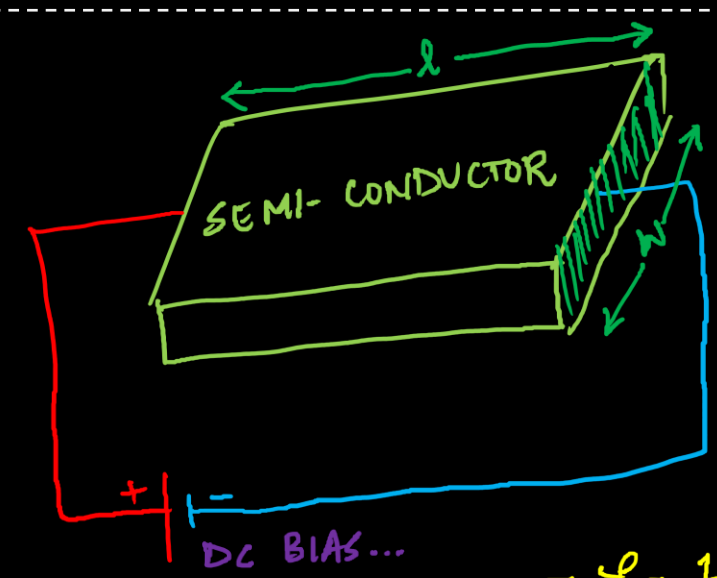
$\sigma = n \cdot e \cdot \mu_e$

For Metals — (1) Unipolar conduction.

Conductivity. n No. of e^- per unit volume. μ_e Mobility of e^- .

$\sigma = n \cdot e \cdot \mu_e + p \cdot e \cdot \mu_h$

Bipolar conduction



Resistance $R = \frac{l}{\sigma \cdot A} = \frac{1}{\sigma} \cdot \frac{l}{A}$

$\sigma = \frac{1}{\rho}$ Conductivity.

$R = \frac{1}{n \cdot e \cdot \mu_e + p \cdot e \cdot \mu_h} \cdot \frac{l}{A}$

Concept of EXTRINSIC SEMICONDUCTORS

EXTRINSIC SEMICONDUCTOR [N - TYPE]

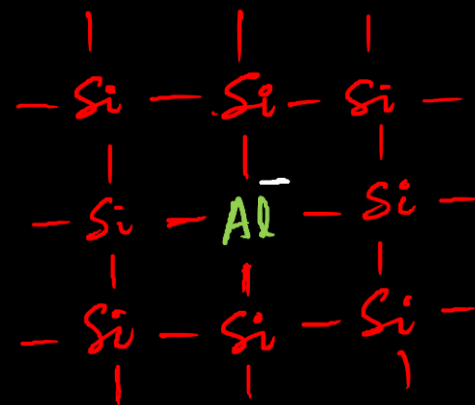
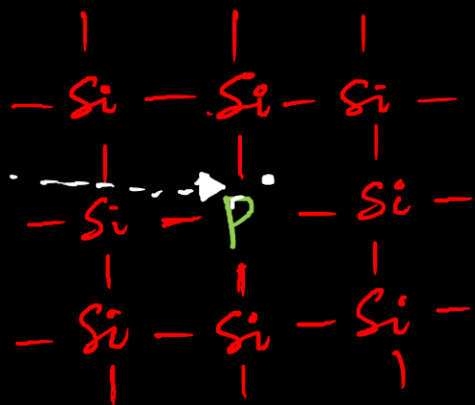
→ Intrinsic + DOPING impurities

N-TYPE

- Pentavalent impurities
- group 15: **As, P** elements

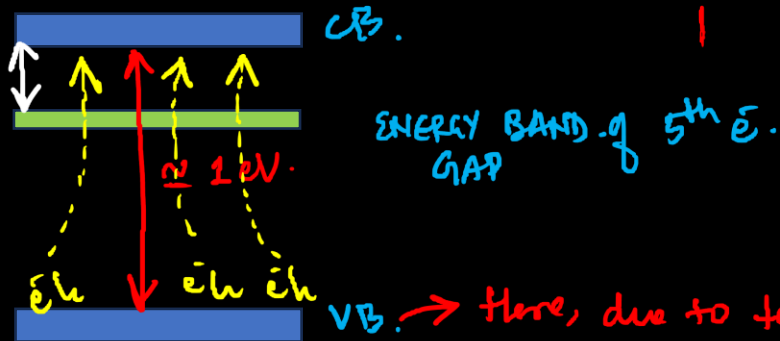
P-TYPE

- Trivalent impurities
- group 13: **B, Al** elements



'P' has valency 5, so it can make bonds with 5 elements but single bond σ. NOT π.
 But this e (•) is the NON-BONDING e of 'P'.
 This e has energy just less than CB & around 1 eV energy in b/w VB & CB energy band gap.

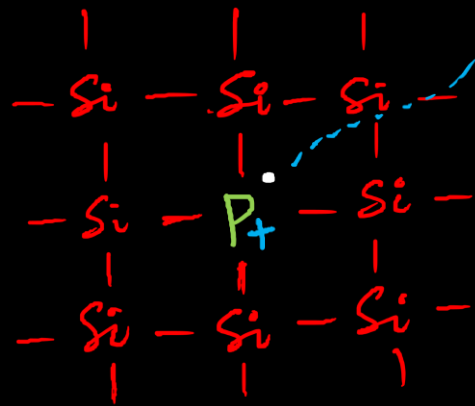
100 times smaller than Eg.



Then, these e's & holes will move to CB.

→ Here, due to temp ≠ 0K; due to vibrational Thermal Energy

CBs will Break → e-h pairs → e } will be generated
 → e-h pairs → h }



Now: After that; if this e^- is removed ...

A e^- deficient vacancy will be created ... i.e. a five charge.

Hence; this phenomenon is a DONOR LEVEL

IONIZATION

process.

Immobile donor ion.

Temperature controlled
phenomenon

or

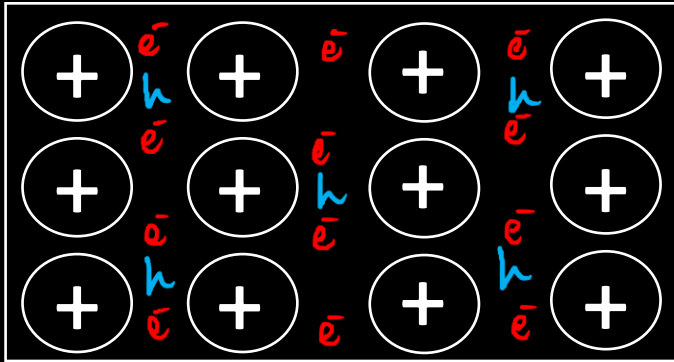
Temperature Sensitive
process

CONCEPT of EXTRINSIC SEMICONDUCTOR

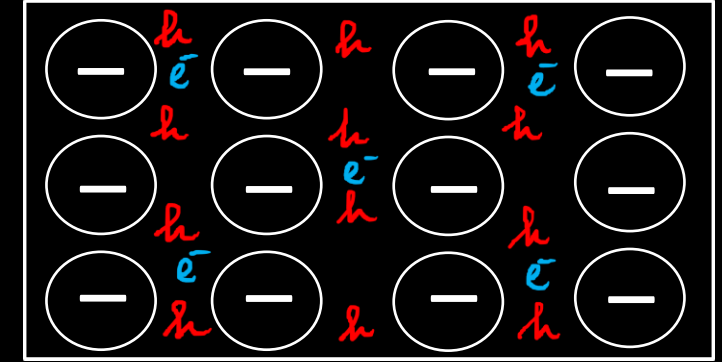
[06-02-2025]

Sailesh Mishra

Representation of N – TYPE s/c



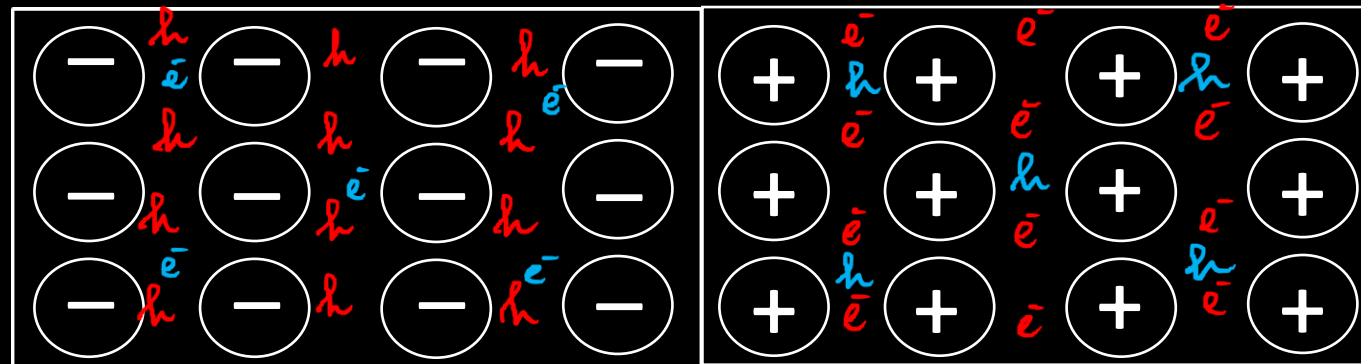
Representation of P – TYPE s/c



+ = Immobile DONOR ions .

- = Immobile ACCEPTOR ions .

e-s & h = MOBILE .



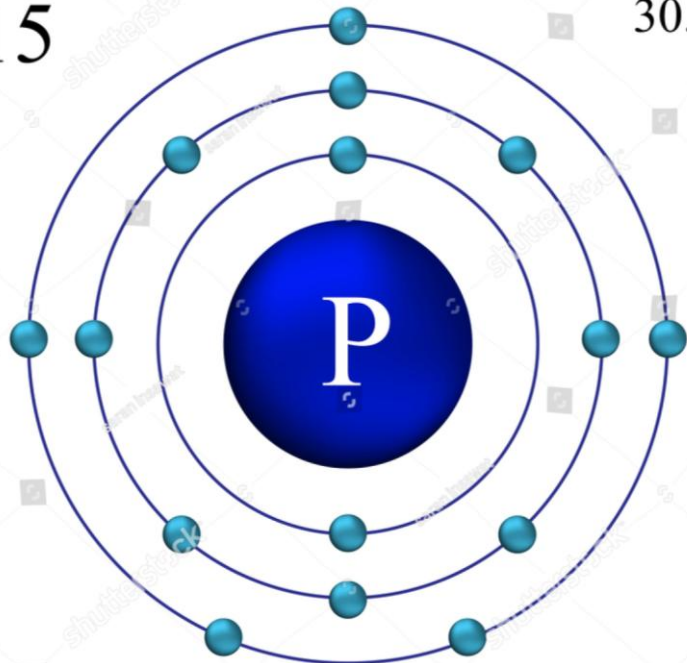
P – N Junction

Atomic number

15

Atomic mass

30.974



Phosphorus

Electron configuration : $1s^2. 2s^2. 2p^6. 3s^2. 3p^3$

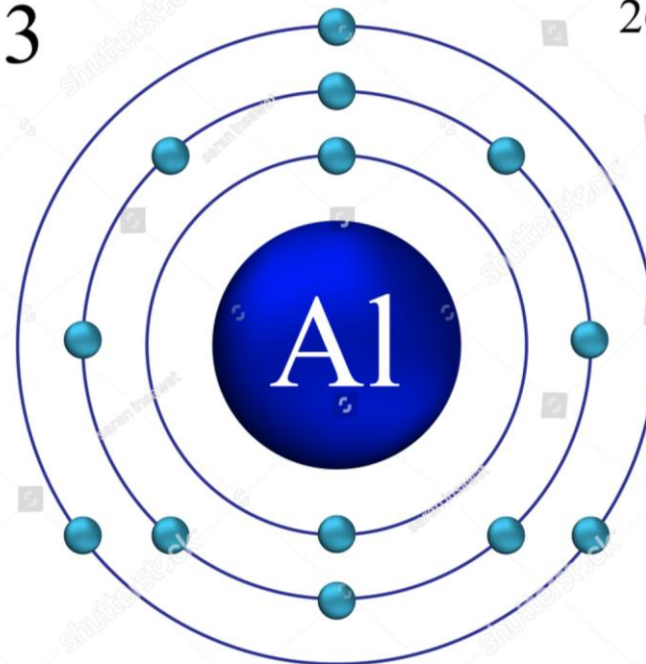
Energy levels : 2.8.5

Atomic number

13

Atomic mass

26.982



Aluminum

Electron configuration : $1s^2. 2s^2. 2p^6. 3s^2. 3p^1$

Energy levels : 2.8.3

Recall from Fig. 2.2 that **phosphorus (P)** contains **five valence electrons**. What happens if some P atoms are introduced in a silicon crystal? As illustrated in Fig. 2.5, each P atom shares four electrons with the neighboring silicon atoms, leaving the fifth electron "unattached." This electron is free to move, serving as a charge carrier. Thus, if N phosphorus atoms are uniformly introduced in each cubic centimeter of a silicon crystal, then the density of free electrons rises by the same amount.



$n_i^2 = \text{intrinsic carrier conc.}$

$$= 1.45 \times 10^3 / \text{cm}^3$$

Figure 2.5 Loosely-attached electron with phosphorus doping.

The controlled addition of an "impurity" such as phosphorus to an intrinsic semiconductor is called "doping," and phosphorus itself a "dopant." Providing many more free electrons than in the intrinsic state, the doped silicon crystal is now called "extrinsic," more specifically, an " **n -type**" semiconductor to emphasize the abundance of free electrons.

As remarked earlier, the electron and hole densities in an intrinsic semiconductor are equal. But, how about these densities in a doped material? It can be proved that even in this case,

$$np = n_i^2,$$

→ **LAW OF MASS ACTION** (2.5)

where n and p respectively denote the electron and hole densities in the extrinsic semiconductor. The quantity n_i represents the densities in the intrinsic semiconductor (hence the subscript i) and is therefore independent of the doping level [e.g., Eq. (2.1) for silicon].

DOPING
%
DOPANT
=

Let us formulate our results thus far. If an ~~intrinsic~~ ^{eg:- Silicon.} semiconductor is doped with a density of N_D ($\gg n_i$) donor atoms per cubic centimeter, then the mobile charge densities are given by

No. of DONOR Atoms

eg:- N-TYPE

Pentavalent

P, As, Sb, etc...

$$\text{Majority Carriers: } n \approx N_D \quad (2.9)$$

$$\text{Minority Carriers: } p \approx \frac{n_i^2}{N_D} \quad (2.10)$$

In N-side

Similarly, for a density of N_A ($\gg n_i$) acceptor atoms per cubic centimeter:

$$\text{Majority Carriers: } p \approx N_A \quad (2.11)$$

$$\text{Minority Carriers: } n \approx \frac{n_i^2}{N_A} \quad (2.12)$$

In P-side

Trivalent

impurities/dopants: B, Al.

Since typical doping densities fall in the range of 10^{15} to 10^{18} atoms/cm³, the above expressions are quite accurate.

EXTRINSIC SEMICONDUCTOR [N - TYPE]

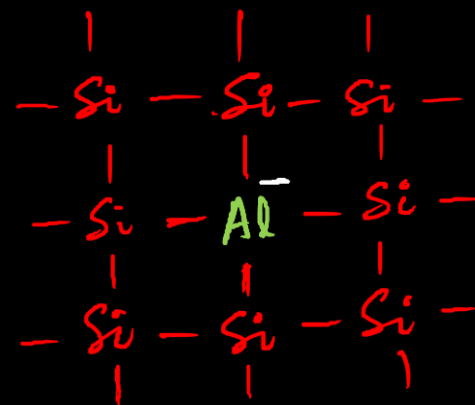
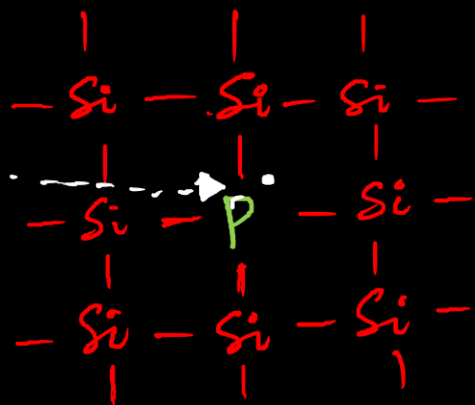
→ Intrinsic + DOPING impurities

N-TYPE

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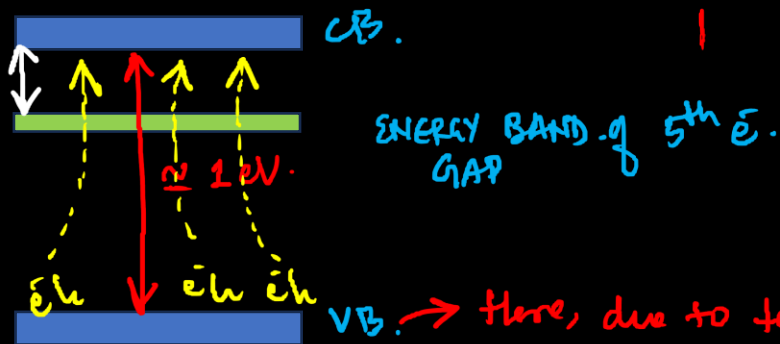
P-TYPE

- Trivalent impurities =
- group 13: **B, Al** elements



'P' has valency 5, so it can make bonds with 5 elements but single bond 5. NOT 4. But this 5th e⁻ is the NON-BONDING e⁻ of 'P'. This e⁻ has energy just less than CB & around 1 eV energy in b/w VB & CB energy band gap.

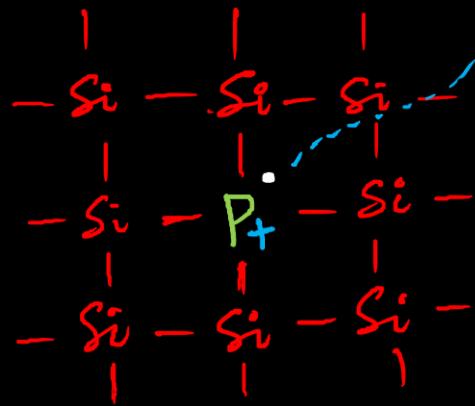
100 times smaller than Eg. =



↑ ↑ ↑
eh eh eh
Then, these e⁻ & holes will move to CB.

→ Here, due to temp ≠ 0K; due to vibrational Thermal Energy

CBs will Break → e-h pairs → e⁻ h⁺ } will be generated =



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Hence; this phenomenon is a DONOR LEVEL

IONIZATION

process.

Immobile donor ion.

Temperature controlled
phenomenon

or

Temperature Sensitive
process

SEMI-CONDUCTOR

INSULATORS < Conductivity < CONDUCTORS

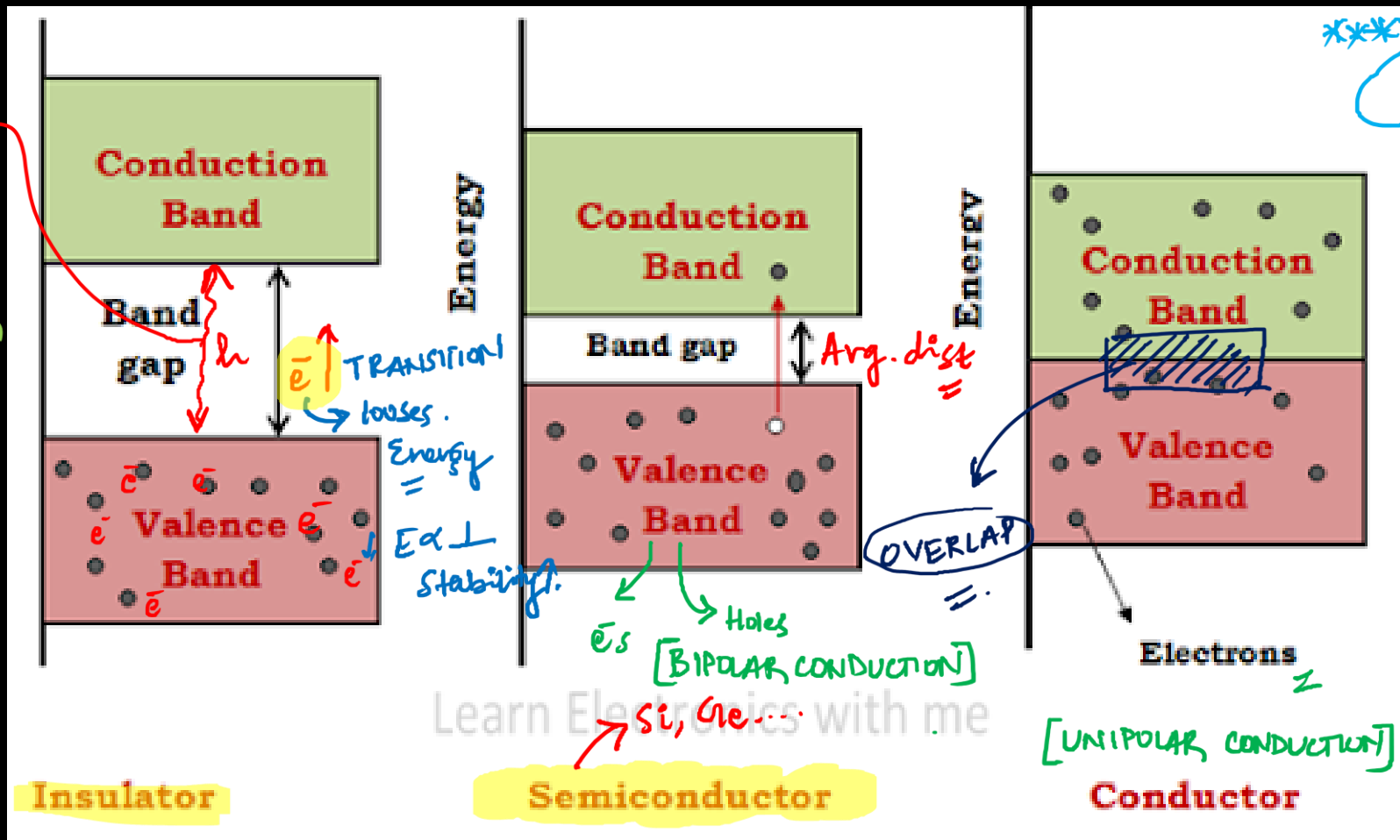
∴ This h is a distance, is a Material dependent property !!

of $h \uparrow$ PE $\uparrow$ TE $\uparrow$ (E $\uparrow$, stability $\downarrow$)

dist $\uparrow$ E $\uparrow$ Cond. $\downarrow$

$$E_g = \frac{1.38 \times 10^{-24} \text{ J/K} \times 300 \text{ K}}{1.6 \times 10^{-19}}$$

$$E_g = 2.5875 \times 10^{-3} \approx 25.87 \text{ mV} \approx \boxed{26 \text{ mV}}$$



DRIFT VELOCITY = $\frac{1}{2} m v_d^2$

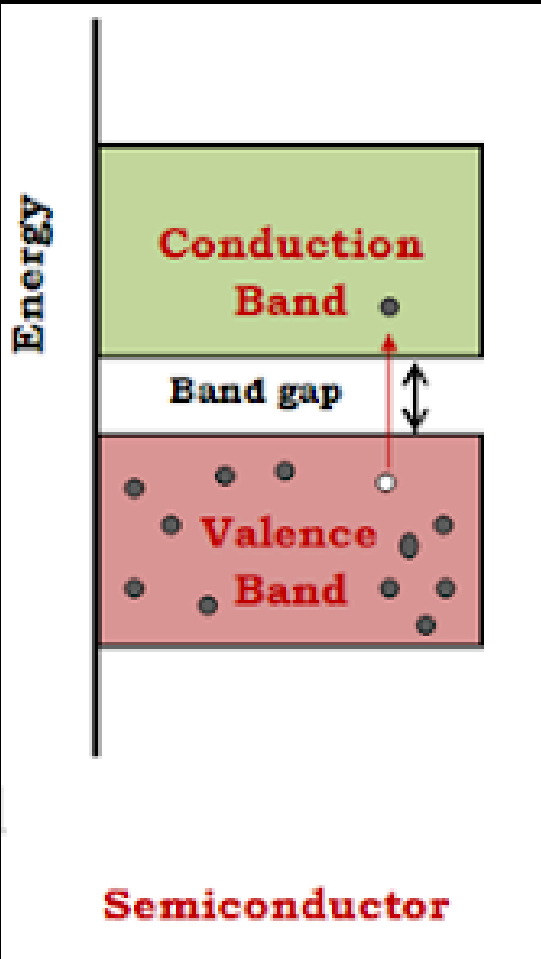
distance $\uparrow$ TE = PE + KE $\uparrow$ mgh
The Energy required to Make a transition from VB $\rightarrow$ CB is w/as BGEL.
[Band Gap Energy level]

• It is directly proportional to $\frac{k_B T}{2}$

where $K_b \rightarrow$ Boltzman Constant, $T =$ Absolute Temp. = $27^\circ\text{C} = 273 + 27^\circ\text{C} = 300\text{K}$.
 $= 1.38 \times 10^{-24} \text{ J/K}$ $2 -$ charge of $e^- = 1.6 \times 10^{-19} \text{ C}$

Learn Electronics with me

Si, Ge...



1. Greater E_g translates to fewer electrons, but a higher T yields more electrons

(Band)

① C.B.

V.B.

Eg: Gap b/w VB & CB.

Gap ① e^- availability will be less.

as dist is more; there is a less chance of getting e^- , who will travel this much distance?

② $n \propto e^{-\frac{E_g}{kT}}$

$\ln(N) = -\frac{E_g}{kT}$

due to the Negative charge ...

$N_e \uparrow$ when $T \uparrow$

No. of e^-

But; $E_g \uparrow$; $N_e \uparrow$

$[N_e: \text{No. of } e^-] =$

to fewer electrons, but a higher T yields more

consider the density (or concentration) of electrons per unit volume, n_i , and write for silicon:

$$n_i = 5.2 \times 10^{15} T^{3/2} \exp \left(\frac{-E_g}{2kT} \right)$$

where $k = 1.38 \times 10^{-23} \text{ J/K}$ is called the Boltzmann constant.

in books on semiconductor physics, e.g., [1]. Insulators display a smaller n_i . Also, as $T \rightarrow 0$ so do $T^{3/2}$ toward zero.

The exponential dependence of n_i upon E_g explains why diamond, which has a large E_g , is an insulator. Conductors, on the other hand, have a moderate E_g , typically ranging from 1 eV to 2 eV.

Example 2.1 Determine the density of electrons in silicon at $T = 600 \text{ K}$.

Solution Since $E_g = 1.12 \text{ eV} = 1.792 \times 10^{-19} \text{ J}$, we have

$$n_i(T = 300 \text{ K}) = 1.08 \times 10^{16} \text{ cm}^{-3}$$

$$n_i(T = 600 \text{ K}) = 1.54 \times 10^{17} \text{ cm}^{-3}$$

Since for each free electron, a hole is left behind, $n_i = p_i$.

Exercise Repeat the above exercise for a material having a band gap of 1.5 eV.

The n_i values obtained in the above example for silicon has $5 \times 10^{22} \text{ atoms/cm}^3$, we recognize that only a small fraction of the atoms are ionized.

³The unit eV (electron volt) represents the energy necessary to move an electron through a potential difference of 1 V. Note that $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$.

$$n_i = 5.2 \times 10^{15} T^{3/2} \exp \left(\frac{-E_g}{2kT} \right) \text{ electrons/cm}^3$$

2. How can np remain constant while we add more donor atoms and increase n ??

Why Does np Remain Constant?

When donor atoms are added (n-type doping), the free electron concentration (n) increases. According to the mass action law, to maintain the constant product $np = n_i^2$, the hole concentration (p) must decrease.

How It Happens:

- Doping introduces a large number of electrons, increasing n significantly.
- Simultaneously, the equilibrium conditions force the hole concentration p to drop proportionally.

This is why:

$$n_{\text{increased}} \cdot p_{\text{reduced}} = n_i^2$$

For example: If n increases by a factor of 1000 due to donor doping, then p must decrease by a factor of 1000 to keep $np = n_i^2$ constant.

Physical Intuition:

- When there are more free electrons in the conduction band, it becomes harder for the valence band to lose electrons (which create holes), reducing the number of available holes.

Thus, the system naturally adjusts to maintain the equilibrium condition $np = n_i^2$ even when donor atoms are added.

Exercise

Why can we not say that $n + p$ should remain constant? *why its $n \cdot p$ (product) = ??*

**Example
2.3**

A piece of crystalline silicon is doped uniformly with phosphorus atoms. The doping density is 10^{16} atoms/cm³. Determine the electron and hole densities in this material at the room temperature.

Solution

The addition of 10^{16} P atoms introduces the same number of free electrons per cubic centimeter. Since this electron density exceeds that calculated in Example 2.1 by six orders of magnitude, we can assume

$$n = 10^{16} \text{ electrons/cm}^3. \quad (2.6)$$

It follows from (2.2) and (2.5) that

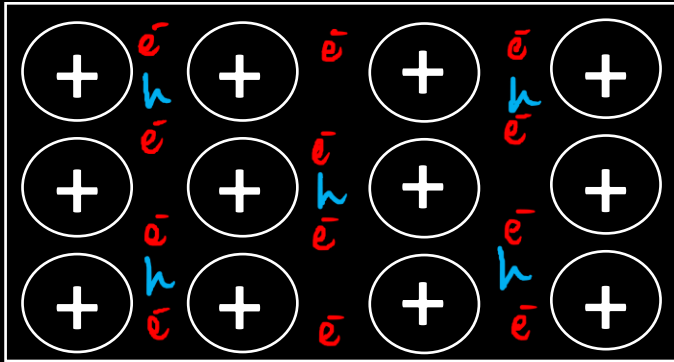
$$p = \frac{n_i^2}{n} \quad (2.7)$$

$$= 1.17 \times 10^4 \text{ holes/cm}^3. \quad (2.8)$$

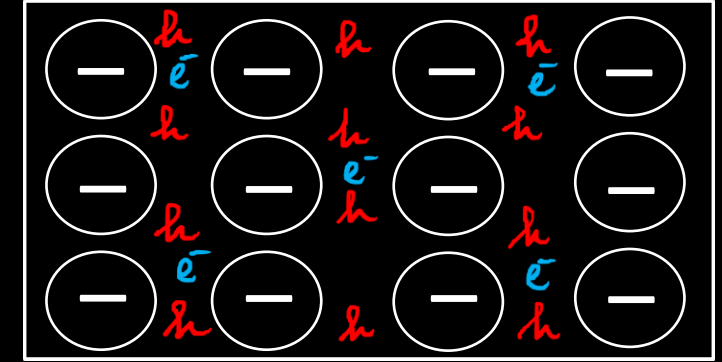
Note that the hole density has dropped below the intrinsic level by six orders of magnitude. Thus, if a voltage is applied across this piece of silicon, the resulting current consists predominantly of electrons.

P-N JUNCTION

Representation of N – TYPE s/c



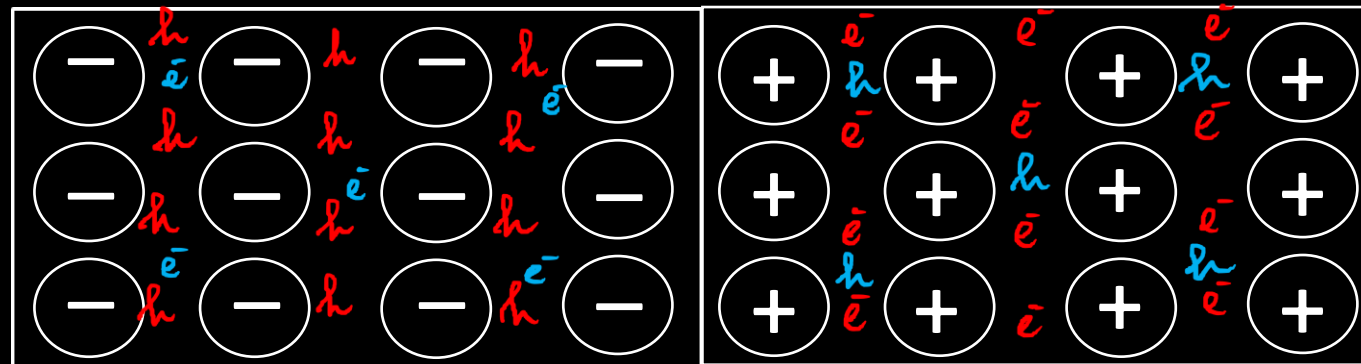
Representation of P – TYPE s/c



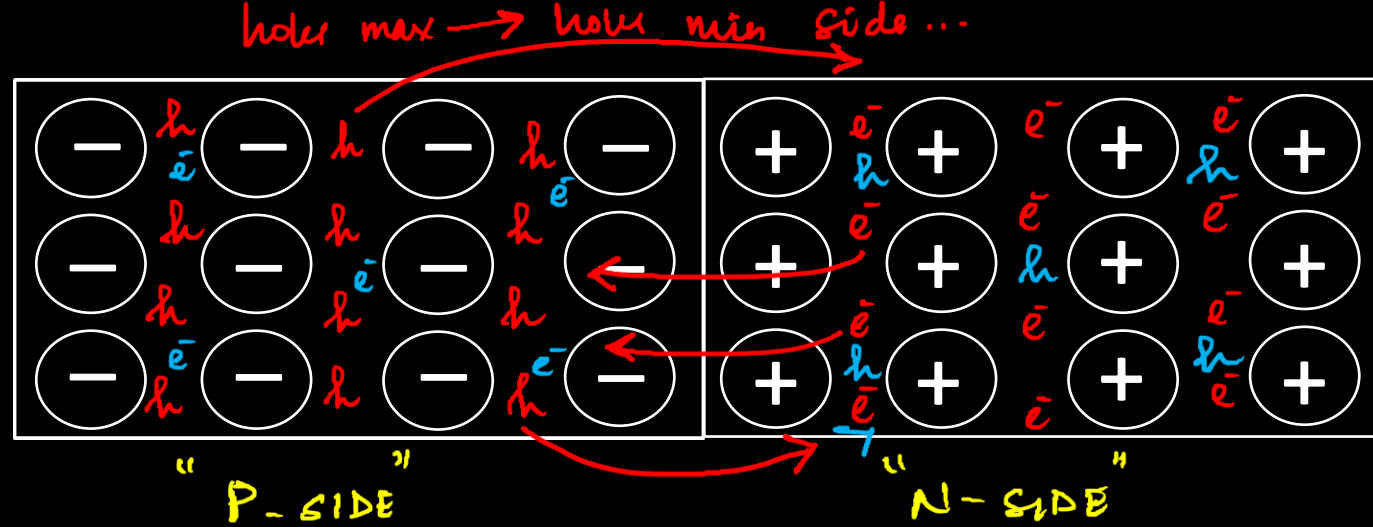
+ = Immobile DONOR ions .

- = Immobile ACCEPTOR ions .

e-s & h = MOBILE CHARGE CARRIERS.



P – N Junction



$\bar{e}s$ - min^m. [holes are max^m] $\xrightarrow{\text{Conc. Gradient}}$ [e⁻s are max^m] holes are min^m

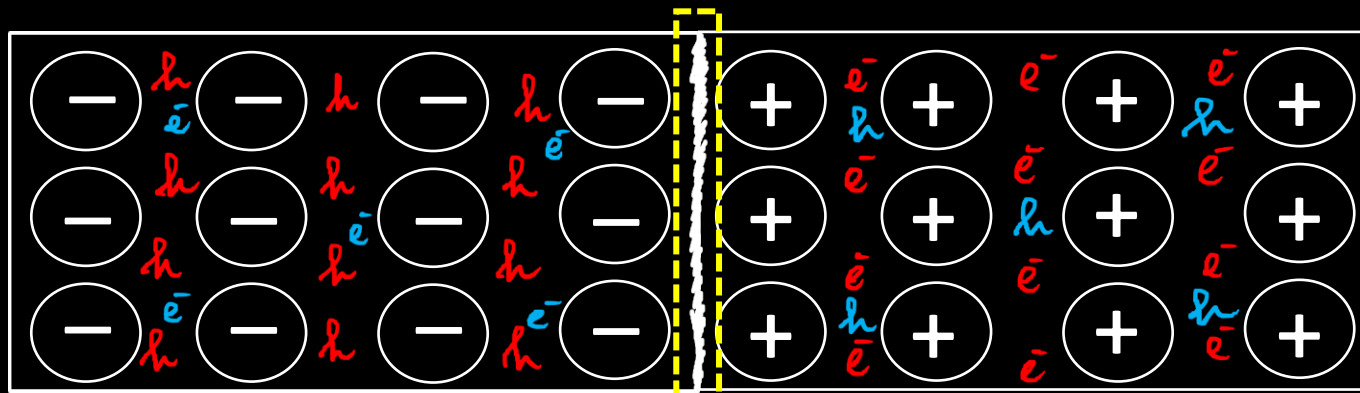
① DIFFUSION

$\rightarrow$ Holes will diffuse from 'P' $\rightarrow$ 'N' side
 $\rightarrow$ e⁻s will diffuse from 'N' $\rightarrow$ 'P' side

② $\bar{e}s$ + holes Recombination.

($\bar{e}-h$) : recombination + creation of CBs [Covalent Bonds].

Once these will form ($\bar{e}h$) pairs, They'll disappear Because they're NOT CARRIERS anymore.



FORMATION OF DIFFUSION CAPACITANCE :- → "P-N" Junction ...

Has Depletion Region / Depletion Width

In order of (μm): Micrometers ...

where; Actually Recombination Occurs ...
of e^- & holes ...

NOW;

P → holes → 'n' side

n → e^- → 'p' side : I ⊕ Because; Current comes from e^- NOT from Holes !!

"BARRIER" POTENTIAL corresponding to the P-N jun"

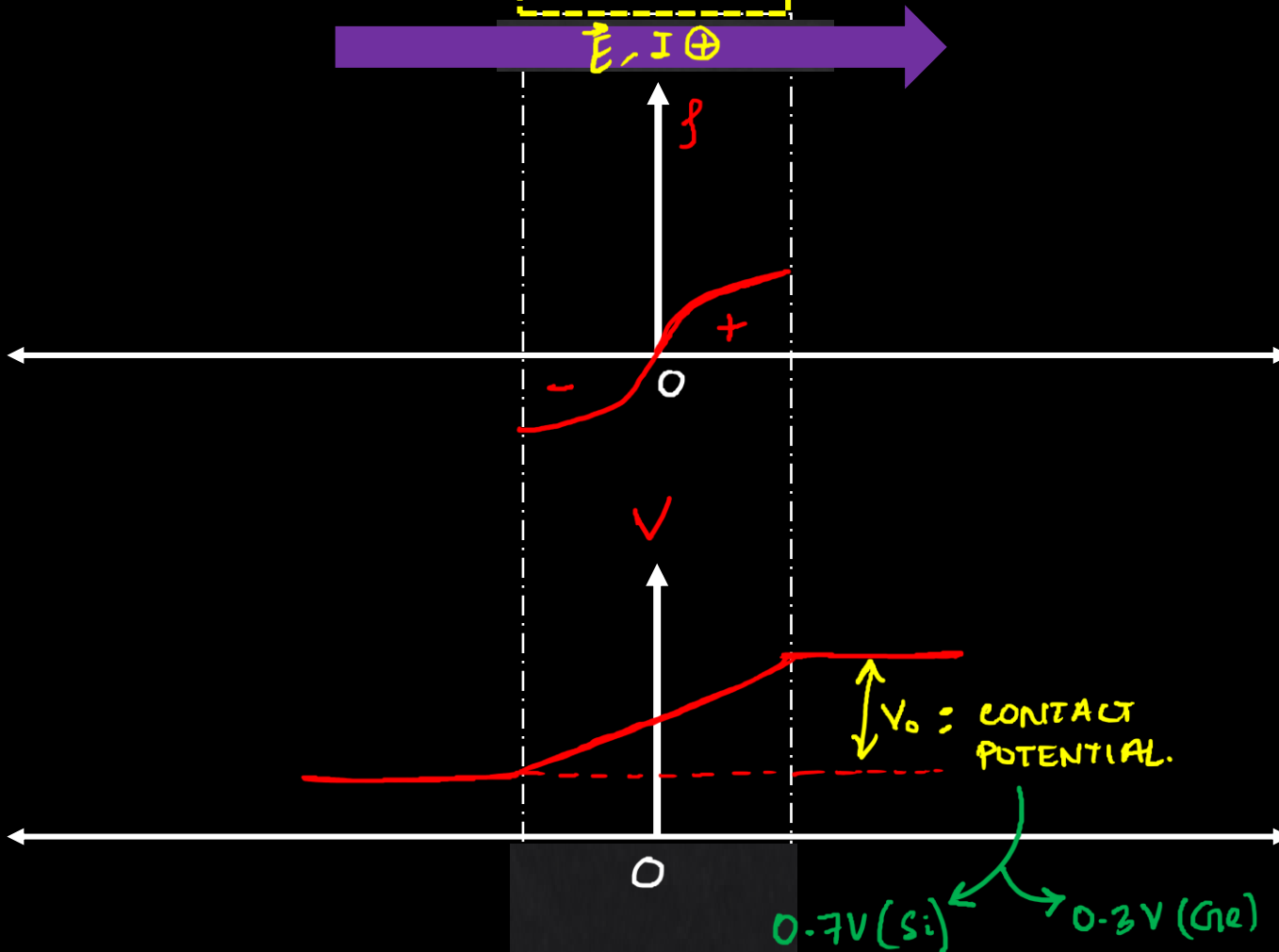
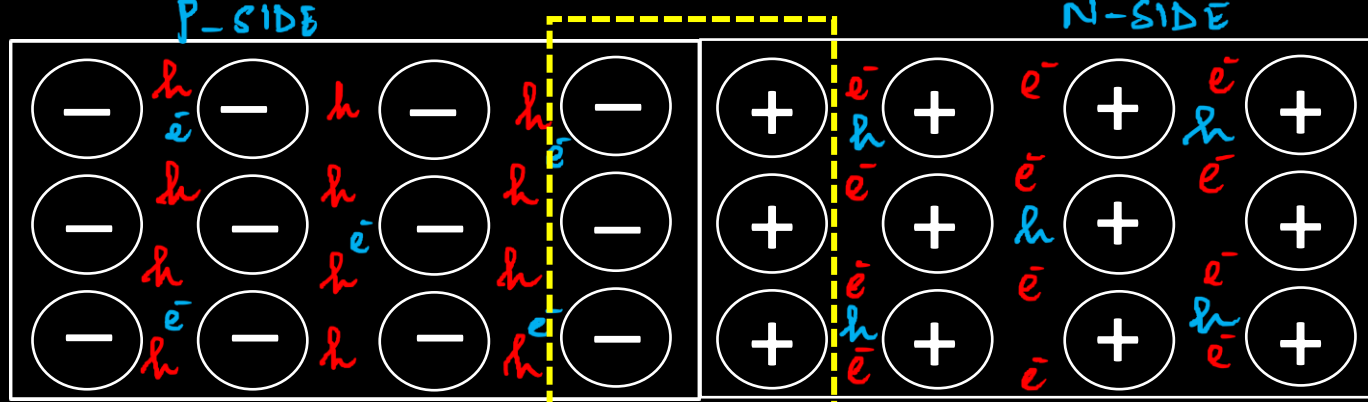
developed ... & an $\vec{E}$ from 'p'-side to 'n'-side too.

1. $C = \frac{\epsilon_0 \cdot A}{d \rightarrow [4\mu m]}$

$C = \frac{\epsilon_0 \cdot A}{d \rightarrow \text{Negligible}}$

$C \propto A'$ → Area of the p-n junction

diffⁿ Capacitance



Let us Assume, N-SIDE potential is higher than P-SIDE By V_0 potential. volts-
=

→ This $\vec{E}$ from P-SIDE to N-side corresponding to $\vec{e}$ s flow from N-side to P-SIDE Because $\vec{e}$ s Movement & I direction is opposite to each other...

→ This $\vec{E}$ will exert a Force for the diffusing minority charge carriers on the N-SIDE [holes].

→ This $\vec{E}$ will NOT ONLY oppose the Holes; But also the $\vec{e}$ s [MAJORITY charge carriers] from N-SIDE to P-SIDE.

→ This $\vec{E}$ is NOTHING But the BARRIER POTENTIAL

→ But this $\vec{E}$ supports the Minority charge Carriers i.e [holes] and the corresponding current -
that the movement of Holes is termed as DRIFT CURRENT ✓.

↳ This is that THERMALLY Generated
Current.
=

As; these Are made/aroused due
to Breaking of CB's.

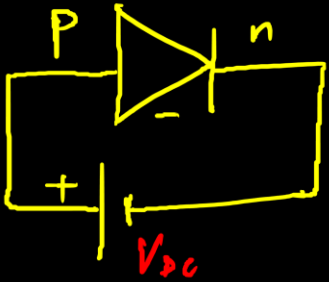
- Followed By:- (e^-h) pair creation...

DC BIASING A DIODE

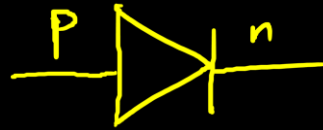
BIASING THE P-N Junction with DC VOLTAGE

Turning ON/applying a Relative Potential difference across the terminals of a P-N jun.
[D.C.]

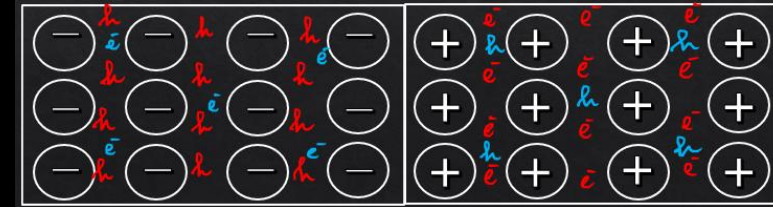
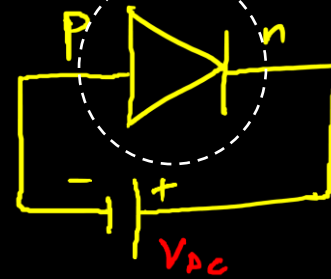
FORWARD
BIASING



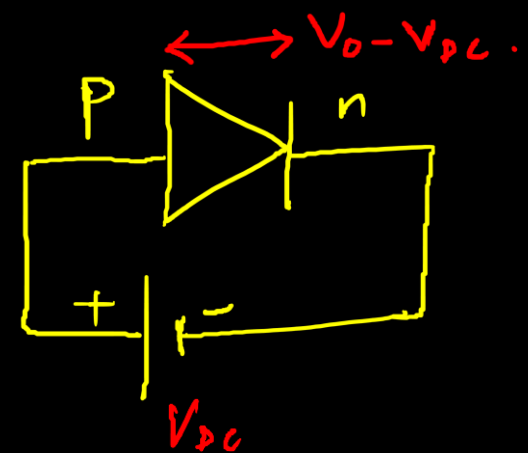
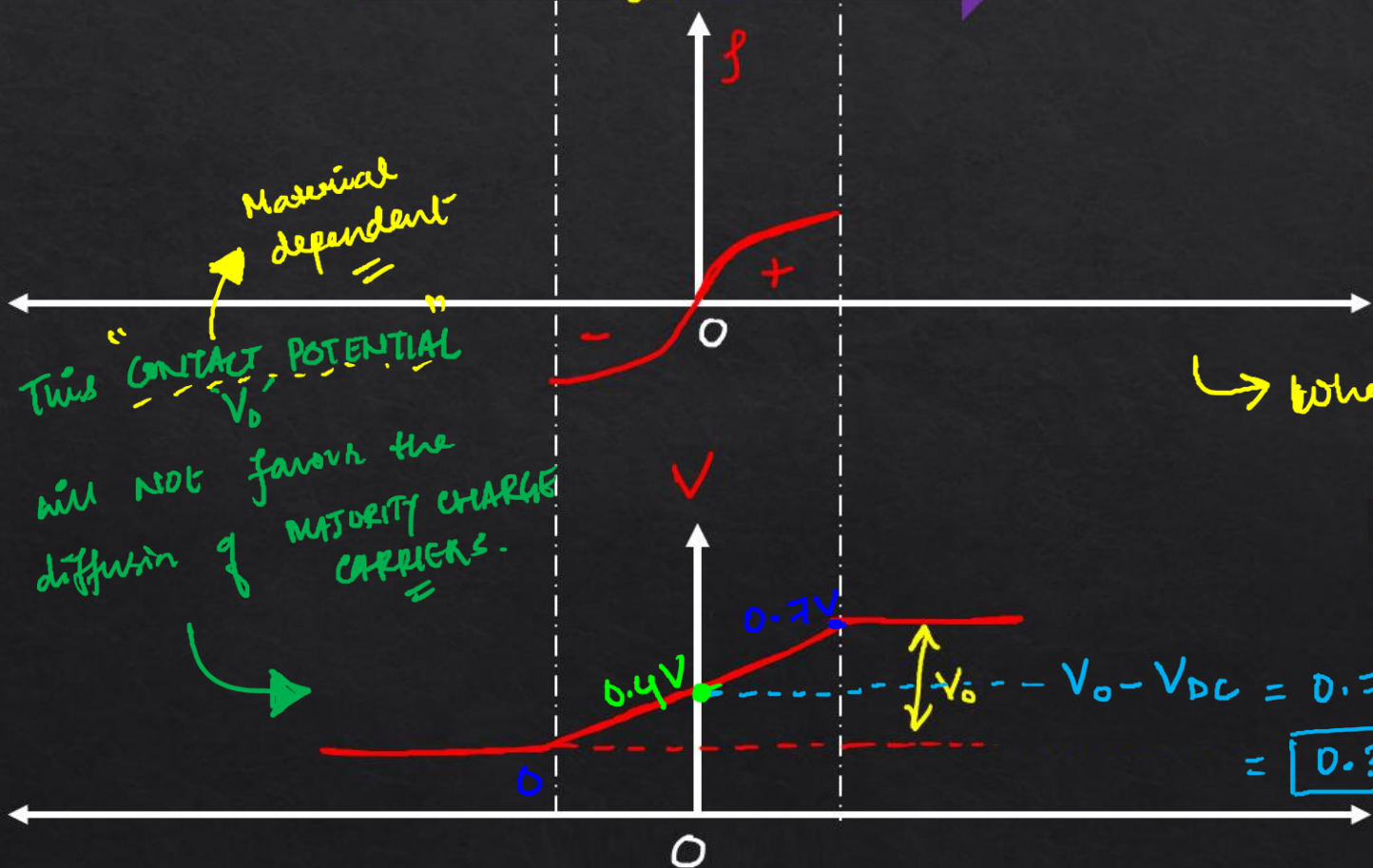
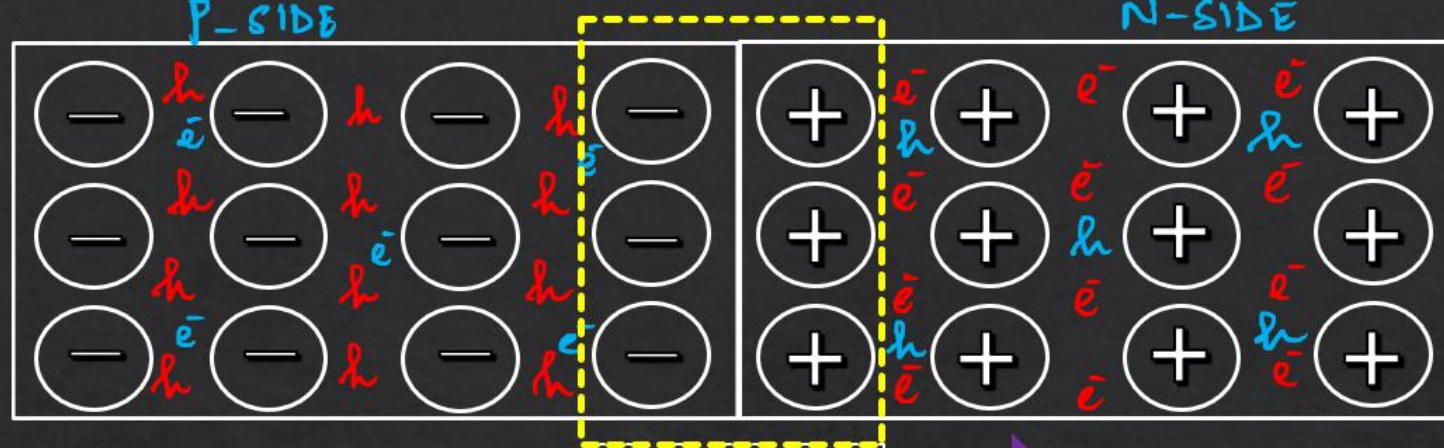
NO
BIASING



REVERSE
BIASING



[P-N Junction]
=



Now, the potential barrier across the P-N junction will be " $V_0 - V_{DC}$ ".

← Contact potential

↓ DC Bias Voltage.

When $V_{DC} > V_0$: P-high ; N-low.

$V_{DC} = V_0$: 0

$V_{DC} < V_0$: P-low ; N-High.

↳ That Means ; if $V_{DC} > V_0$; Contact potential / V_T will become lower and conduction of e^- & Holes will START !! .

In FB ; $V \oplus$, $I \oplus$: DRIFT + DIFFUSION Both current starts to Flow .

$$I_{NET} = \underset{\uparrow}{I_{diff}} - \underset{\downarrow}{I_{drift}}$$

In RB ; $V \ominus$; $I \ominus$: Only DRIFT current Flows .

Both Drift Current and Diffusion Current Can Involve Electrons and Holes:

- **Drift Current:** This current arises due to the presence of an **external electric field** applied to the PN junction. This field exerts a force on both **electrons (in the n-region)** and **holes (in the p-region)**, causing them to **drift** in the direction opposite to the field (electrons move towards the positive terminal and holes towards the negative terminal).
- **Diffusion Current:** This current results from the tendency of charge carriers to move from regions of **high concentration** to regions of **low concentration**. In an unbiased PN junction, there's a concentration gradient across the depletion region:
 - **Electrons** have a higher concentration in the n-region and a lower concentration in the p-region.
 - **Holes** have a higher concentration in the p-region and a lower concentration in the n-region.

This diffusion process creates a **diffusion current of electrons** flowing from the n-region to the p-region and **holes** flowing from the p-region to the n-region.

In forward bias, both drift and diffusion currents contribute to the overall current flow in a PN junction diode, while in reverse bias, only a very small drift current is present.

Forward Bias :

1. Diffusion Current Dominates:

- When a forward bias is applied (positive to p-region, negative to n-region), the depletion region narrows. This allows for a significant concentration gradient of charge carriers across the junction:
 - Electrons are abundant in the n-region and scarce in the p-region.
 - Holes are abundant in the p-region and scarce in the n-region.
- This strong concentration gradient drives a large **diffusion current** of electrons flowing from n to p and holes flowing from p to n.

2. Drift Current Assists:

- The forward bias also creates an **external electric field** that points from the positive terminal (p-region) towards the negative terminal (n-region).
- This electric field exerts a force on both electrons and holes, causing them to **drift** in the opposite direction of the field (electrons move towards n and holes towards p).
- While **diffusion current** is the dominant factor due to the significant concentration gradient, the **drift current** adds to the overall flow of charge carriers in the same direction.

Reverse Bias :

1. Limited Diffusion:

- When a reverse bias is applied (positive to n-region, negative to p-region), the depletion region widens significantly.
- This creates a larger barrier that severely restricts the movement of charge carriers across the junction by diffusion.
- The concentration gradient remains, but the depletion region acts as a formidable obstacle, leading to a negligible diffusion current.

2. Small Drift Current:

- The reverse bias creates an **external electric field** that points from the positive terminal (n-region) towards the negative terminal (p-region).
- This electric field does accelerate a small number of **minority carriers** (electrons in the p-region and holes in the n-region) in the direction of the field. However, these minority carriers are very few in number, leading to a very small **drift current** in the opposite direction of the applied voltage. This is known as the **leakage current**.

In essence, the strong forward bias allows both diffusion and drift currents to flow, while the reverse bias drastically limits diffusion and allows only a minuscule drift current due to minority carriers.